Quantum Computers

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(Winter semester 2025.)

Quantum Computers (undergraduate/graduate course)

Course introduced in 2008 with the intention to present elementary concepts of quantum mechanics (QM) in a more modern setting than that of a traditional introductory quantum physics course.

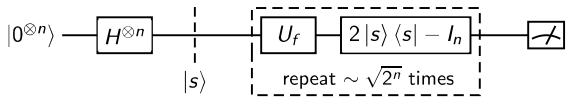
Main learning outcomes:

- Master **concepts of QM** (superposition, entanglement) in finite dimensional Hilbert spaces.
- Be able to read **quantum logical circuits** and understand simple quantum algorithms (Deutsch, Grover).

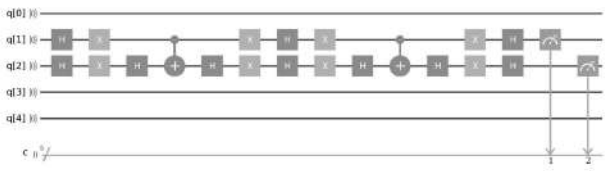
Literature:

- M. Le Bellac, *A Short Introduction to Quantum Information and Quantum Computing*, Cambridge University Press, 2006.

Example: Grover's search algorithm logical circuit:

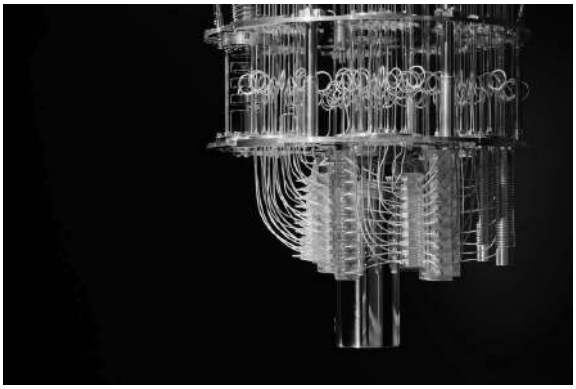


Implementation in IBM Q Composer: $n = 2$, $|w\rangle = |00\rangle$



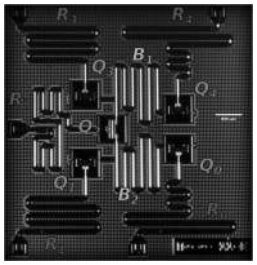
http://quantum.ibm.com

A quantum computer?



http://quantum.ibm.com

2017: Real 5-qubit chips were made available to the public by IBM. Real quantum computations (experiments) could be conducted from classrooms at FER.



http://quantum.ibm.com

2023: IBM announced withdrawal of 5-qubit chips. 127-qubit to become available soon.

Outline of the first half of the course:

- Introduction to quantum phenomena (polarisation of light)
- Quantum key distribution protocol BB84
- Principles of quantum mechanics
- Representation of quantum bit state using Bloch's sphere
- Entanglement in two-qubit systems
- Realisation of quantum bit using spin 1/2
- State operator (density matrix) formalism

Outline of the second half of the course (working on it):

- Quantum information properties
- Quantum logical circuit
- Simple communication protocols
- Algorithms: Deutsch, Grover, Quantum Fourier Transform
- Error correction codes

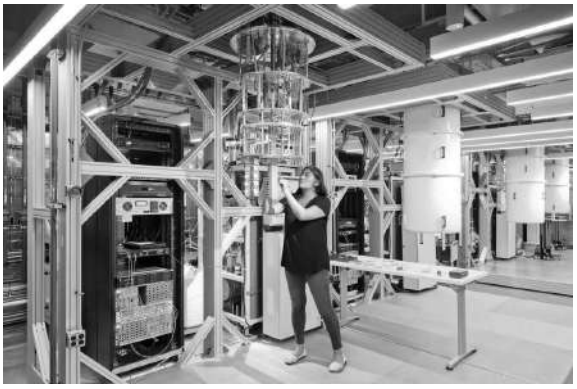
Some points on the quantum physics and quantum computing timeline:

- 1927: Heisenberg's uncertainty principle formulated
  https://doi.org/10.1007/BF01397280
- 1935: Entanglement in quantum systems realized (EPR paradox)
  https://doi.org/10.1103/PhysRev.47.777
  https://doi.org/10.1017/s0305004100013554
- 1964: Major step in understanding of quantum correlations (Bell's inequalities)
  https://doi.org/10.1103/PhysicsPhysiqueFizika.1.195
- 1981: Richard Feynman advocates the use of quantum computers in attempts to simmulate physical systems
  https://doi.org/10.1007/BF02650179

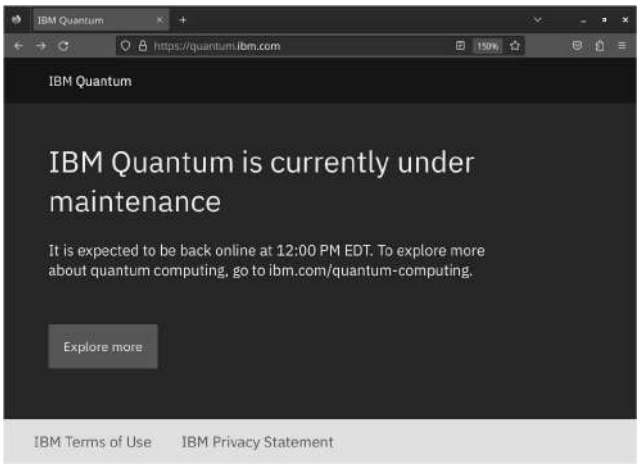
- 1982: Non-cloning theorem formulated
  https://doi.org/10.1038/299802a0
  https://doi.org/10.1016/0375-9601(82)90084-6
- 1984: Quantum key distribution protocol BB84
  https://doi.org/10.1016/j.tcs.2014.05.025
- 1985: Deutsch's algorithm
  https://doi.org/10.1098/rspa.1985.0070
- 1993: Quantum "teleportation"
  https://doi.org/10.1103/PhysRevLett.70.1895
- 1994: Shor's algorithm (factorization, threatens RSA)
  https://doi.org/10.1109/sfcs.1994.365700
- 1996: Grover's search algorithm
  https://doi.org/10.1145/237814.237866

- 2001: Number 15 factored on quantum hardware
  https://doi.org/10.1038/414883a

  item 2012: Number 21 factored using Shor's algorithm
  https://doi.org/10.1038/nphoton.2012.259
- 2012: First "transmon" quantum chips become operational
- 2017: IBM makes 5-qubit quantum chips available to general public
- 2019: Attempt to factor number 35 on IBM's first commercial quantum computer "Q System One" failed
  https://doi.org/10.1103/PhysRevA.100.012305
- 2019: Google declares quantum supremacy (IBM disputes)
  https://doi.org/10.1038/s41586-019-1666-5



http://quantum.ibm.com



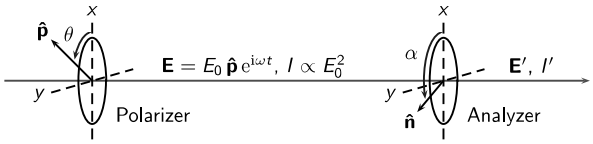
Introduction to quantum phenomena

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In classical physics, a beam of light is described as an *electromagnetic wave.*

The wave can be polarized in a variety of ways. Having passed through a polarizer, it is *linearily polarized:*



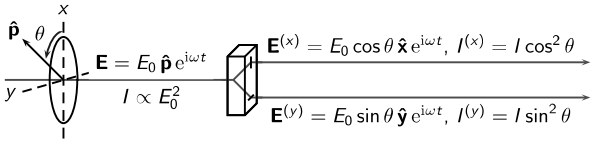
Having passed through another polarizer (aka analyzer):

$$\mathbf{E}' = (\mathbf{E}\cdot\hat{\mathbf{n}})\,\hat{\mathbf{n}} = E_0(\hat{\mathbf{p}}\cdot\hat{\mathbf{n}})\,\hat{\mathbf{n}}\,e^{i\omega t} = E_0\cos[\theta-\alpha]\,\hat{\mathbf{n}}\,e^{i\omega t}$$

$$\Longrightarrow \qquad I' = I\cos^2[\theta-\alpha] \qquad \text{(The law of Malus)}$$

Passing through a birefringent plate, any beam decomposes into two parallel beams with mutually perpendicular linear polarizations.

Here we decompose a linearily polarized beam:



Output beam intensities obey Malus's law; Energy is conserved:

$$I^{(x)} + I^{(y)} = I$$

The experiment can be arranged so that the two output beams merge into one, restoring the original wave:

$$\mathbf{E}^{(x)} + \mathbf{E}^{(y)} = E_0(\cos\theta\,\hat{\mathbf{x}} + \sin\theta\,\hat{\mathbf{y}})e^{i\omega t} = E_0\,\hat{\mathbf{p}}\,e^{i\omega t} = \mathbf{E}.$$

In quantum physics, a beam of light is understood as a stream of particle-like photons.

A detector (aka photon-counter) can detect only a whole number (integer) of photons $\mathcal{N}$:

$$\mathcal{N} = 0, 1, 2, \ldots$$

The intensity of a beam of light is proportional to the number of photons that in unit time pass through some surface (eg. enter a photon-counter):

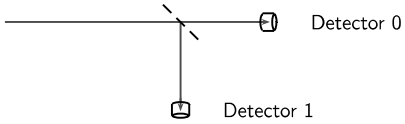
$$I \propto \frac{d\mathcal{N}}{dt}$$

Remarkable properties of phenomena "at quantum level" are:

- Experiments exist in which the final outcome can not be predicted with certainty; Only the probabilities of possible outcomes can be predicted.
- Experiments exist in which it is *not even in principle* possible to determine in which of several ways certain outcome has been reached.

In quantum optics, experiments in which it is not even in principle possible to determine along which of several paths a photon has reached a detector are colloquially known as "which path" experiments.

Example: Half-transparent mirror and a pair of detectors as a random number generator



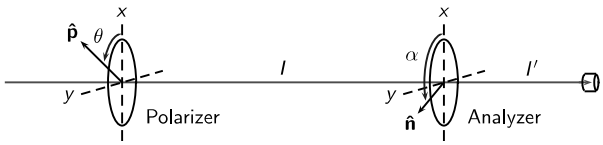
It is *not even in principle* possible do predict in which detector the photon will end.

Detections at detectors 0 and 1 generate a random sequence

$$\ldots, 0, 1, 0, 0, 1, 1, 1, 0, 1, 1, 1, 0, 1, 1, 0, 0, 0, 1, 0, 0, \ldots$$

that can be used as a true random number generator.

Example: Photons of a linearly polarized beam passing through an analyzer or being absorbed by it

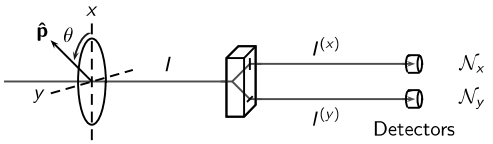


It is *not even in principle* possible to predict if a photon will pass through the analyzer or will be absorbed (except for $\theta - \alpha = 0, \pm\frac{\pi}{2}, \pm\pi, \dots$).

The probability that a photon of a linearly polarized beam (orientation θ) passes through the analyzer (orientation α) follows from Malus's law as

$$p(\theta \to \alpha) = \frac{I'}{I} = \cos^2[\theta - \alpha].$$

Example: A polarized beam passing through a birefringent plate



It is *not even in principle* possible to predict in which detector a photon will end (except for $\theta = 0, \pm\frac{\pi}{2}, \pm\pi, \dots$).

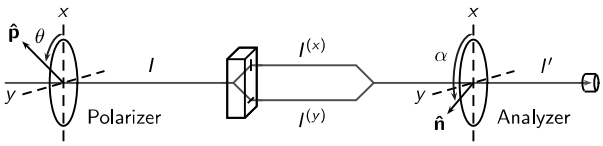
If $\mathcal{N} = \mathcal{N}_x + \mathcal{N}_y$ photons are detected, Malus's law gives:

$$\lim_{\mathcal{N}\to\infty} \frac{\mathcal{N}_x}{\mathcal{N}} = \frac{I^{(x)}}{I} = \cos^2\theta \qquad \lim_{\mathcal{N}\to\infty} \frac{\mathcal{N}_y}{\mathcal{N}} = \frac{I^{(y)}}{I} = \sin^2\theta$$

Probabilities of a photon taking one or the other path follow as:

$$p(\theta \to x) = \cos^2\theta \qquad p(\theta \to y) = \sin^2\theta$$

Example: Experiment in which *not even in principle* one can tell along *which path* a photon reached the detector



Experimentally one finds: $p(\theta \to \alpha) = I'/I = \cos^2[\theta - \alpha]$

However, simple addition of probabilities of taking one path or the other gives a result which is *not in agreement* with experiment:

$$\begin{aligned} p(\theta \to \alpha) &= p(\theta \to x \to \alpha) + p(\theta \to y \to \alpha) \\ &= p(\theta \to x)\, p(x \to \alpha) + p(\theta \to y)\, p(y \to \alpha) \\ &= \cos^2\theta \cos^2\alpha + \sin^2\theta \sin^2\alpha \end{aligned}$$

Probability amplitude

In quantum mechanics, to a transition of a physical system from state α to state β there corresponds the *probability amplitude*

$$a(\alpha \to \beta),$$

which is a complex number.

Probability

The *probability* that a system undergoes a transition from state α to state β is the square of the modulus of the corresponding probability amplitude,

$$p(\alpha \to \beta) = \left|a(\alpha \to \beta)\right|^2.$$

Some properties of the probability amplitude:

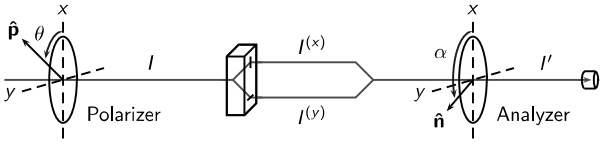
- Probability amplitude for the transition $\alpha \to \beta \to \gamma$:

$$a(\alpha \to \gamma) = a(\alpha \to \beta \to \gamma) = a(\alpha \to \beta)\, a(\beta \to \gamma)$$

- Probability amplitude for a transition that can take place in two ways, $\alpha \to \beta_1 \to \gamma$ and $\alpha \to \beta_2 \to \gamma$, if *not even in principle* one can determine in which of the two ways the transition evolved, is the sum of the two probability amplitudes:

$$\begin{aligned} a(\alpha \to \gamma) &= a(\alpha \to \beta_1 \to \gamma) + a(\alpha \to \beta_2 \to \gamma) \\ &= a(\alpha \to \beta_1)\, a(\beta_1 \to \gamma) + a(\alpha \to \beta_2)\, a(\beta_2 \to \gamma) \end{aligned}$$

Example: *"Which path"* experiment (considered earlier)



If for probability amplitudes of individual transitions one takes

$$\begin{aligned} a(\theta \to x) &= \cos\theta, & a(x \to \alpha) &= \cos\alpha, \\ a(\theta \to y) &= \sin\theta, & a(y \to \alpha) &= \sin\alpha, \end{aligned}$$

the complete probability amplitude becomes

$$\begin{aligned} a(\theta \to \alpha) &= a(\theta \to x)\, a(x \to \alpha) + a(\theta \to y)\, a(y \to \alpha) \\ &= \cos\theta\, \cos\alpha + \sin\theta\, \sin\alpha \\ &= \cos[\theta - \alpha] \qquad \text{(in agreement with experiment).} \end{aligned}$$

The general state of polarization of a monochromatic plain EM wave can be written as the superposition of two waves with mutually perpendicular orientations of linear polarization,

$$\mathbf{E} = E_x\,\hat{\mathbf{x}}\,\mathrm{e}^{\mathrm{i}(\omega t+\phi_x)} + E_y\,\hat{\mathbf{y}}\,\mathrm{e}^{\mathrm{i}(\omega t+\phi_y)} = E_0\,(\lambda\,\hat{\mathbf{x}} + \mu\,\hat{\mathbf{y}})\,\mathrm{e}^{\mathrm{i}\omega t},$$

where $E_0 = \sqrt{E_x^2 + E_y^2}$, and complex numbers

$$\lambda = \frac{E_x}{E_0}\,\mathrm{e}^{\mathrm{i}\phi_x} \qquad \text{and} \qquad \mu = \frac{E_y}{E_0}\,\mathrm{e}^{\mathrm{i}\phi_y}$$

satisfy the condition $|\lambda|^2 + |\mu|^2 = 1$.

It follows that the vector $\lambda\,\hat{\mathbf{x}} + \mu\,\hat{\mathbf{y}}$ encodes the state of polarization of the wave, *or of the photons if quantum picture is used.*

A photon itself can be in any of infinitely many different states of polarization, but an experiment measuring the state of its polarization can have *only two different outcomes* such that the photon can be prepared in a way that one or the other outcome is obtained with certainty.



The experimental device (the birefringent plate) can be "oriented" in different ways, which implies that there are infinitely many different ways of measuring the state of polarization of a photon.

However, *one photon can be measured only once.*

Quantum bit or qubit

Any physical system in which the most general measurement can have only two different outcomes such that the system can be prepared in a way that either of the two is obtained with certainty is called a *quantum bit* or *qubit*

Any physical system qualifying as a qubit is considered the most elementary carrier of quantum information.

Qubit is also the elementary unit of quantum information.

Comparison of classical and quantum bit

Common property of the classical and the quantum bit is that when measured both give one of two different values. These are the differences:

- Classical bit can be in one of its two different states. Quantum bit (until measured) can be in an infinity of different states.
- There is only one way of measuring the state of a classical bit. A quantum bit can be measured in an infinity of different ways.
- The measurement of a classical state leaves the state unchanged. The measurement of a quantum bit in general changes its state.

In computations of probability amplitudes in quantum mechanics it is practical to use the so called bra-ket (or Dirac) notation.

The states of polarization of photons that are linearily polarized with the orientations along the x-axis and along the y-axis we represent with ket-symbols

$$|x\rangle \qquad \text{and} \qquad |y\rangle\,.$$

The general state of polarization of a photon is also represented by ket-symbol which can, in complete analogy with the description of polarization in electromagnetism, be expressed as a superposition

$$|\Phi\rangle = \lambda\,|x\rangle + \mu\,|y\rangle\,, \qquad \lambda,\mu \in \mathbb{C}, \qquad |\lambda|^2 + |\mu|^2 = 1.$$

In the mathematical sense, the ket-symbols such as $|\Phi\rangle$ and $|\Psi\rangle$ represent vectors of the two-dimensional complex *Hilbert* vector space $\mathcal{H}^{(2)}$.

Interior product of vectors $|\Phi\rangle$ i $|\Psi\rangle$ in Hilbert space has the properties

$$\langle\Psi|\Phi\rangle = \langle\Phi|\Psi\rangle^* \in \mathbb{C}, \qquad \langle\Phi|\Phi\rangle \geq 0.$$

The norm of a vector $|\Phi\rangle$ is

$$||\Phi|| = \sqrt{\langle\Phi|\Phi\rangle}.$$

The symbol $\langle\Phi|$ is called the *bra*-symbol, while the object $\langle\Psi|\Phi\rangle$ is known as the *bra-ket*.

If for $|x\rangle$ i $|y\rangle$ in $\mathcal{H}^{(2)}$ it holds

$$\langle x|y\rangle = 0, \qquad \langle x|x\rangle = \langle y|y\rangle = 1,$$

these vectors form an orthonormal basis of $\mathcal{H}^{(2)}$.

If we express vectors $|\Phi\rangle$ i $|\Psi\rangle$ using the basis $\{|x\rangle, |y\rangle\}$ as

$$|\Phi\rangle = \lambda\,|x\rangle + \mu\,|y\rangle\,, \qquad |\Psi\rangle = \nu\,|x\rangle + \sigma\,|y\rangle\,,$$

the interior product of vectors $|\Phi\rangle$ and $|\Psi\rangle$ and the norm of the vector $|\Phi\rangle$ are:

$$\langle\Psi|\Phi\rangle = \nu^*\lambda + \sigma^*\mu = \langle\Phi|\Psi\rangle^* \qquad ||\Phi||^2 = \langle\Phi|\Phi\rangle = |\lambda|^2 + |\mu|^2$$

Photon polarization states are represented by unit-normalized vectors in $\mathcal{H}^{(2)}$. If $|\Phi\rangle$, $|\Psi\rangle$, etc. represent polarisation states,

$$\langle\Phi|\Phi\rangle = 1, \quad \langle\Psi|\Psi\rangle = 1, \quad \ldots$$

The interior product of the states $|\Phi\rangle$ i $|\Psi\rangle$ gives the probability amplitude of the transition of a photon from polarisation state $|\Phi\rangle$ to polarisation state $|\Psi\rangle$.

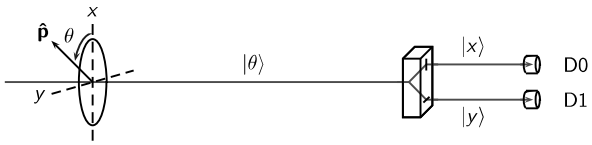
Therefore, the probability amplitude for a photon in the state $|\Phi\rangle$ to be measured (detected) in the state $|\Psi\rangle$ is

$$a(\Phi \to \Psi) = \langle\Psi|\Phi\rangle\,,$$

and the corresponding probability is

$$p(\Phi \to \Psi) = |\,\langle\Psi|\Phi\rangle\,|^2.$$

Example: Measurement of linearily polarized photons



The photons are prepared in the state

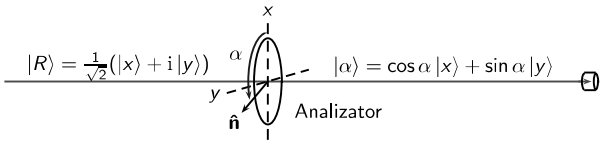
$$|\theta\rangle = \cos\theta\,|x\rangle + \sin\theta\,|y\rangle\,.$$

The probability that a photon is measured in the state $|x\rangle$ (detected in D0) and the probability that it is measured in the state $|y\rangle$ (detected in D1) are

D0: $\quad p(\theta \to x) = |a(\theta \to x)|^2 = |\,\langle x|\theta\rangle\,|^2 = \cos^2\theta$

D1: $\quad p(\theta \to y) = |a(\theta \to y)|^2 = |\,\langle y|\theta\rangle\,|^2 = \sin^2\theta$

Example: Probability of the transition of a photon in the state of right-handed circular polarization $|R\rangle$ to that of linear polarisation $|\alpha\rangle$



$$p(R \to \alpha) = |\,\langle\alpha|R\rangle\,|^2 = \frac{1}{2}\,|\cos\alpha + \mathrm{i}\sin\alpha|^2 = \frac{1}{2}$$

Note that the probability does not depend on α.

Quantum key distribution protocols (BB84)

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rotocols (QKD)

Encripted communication protocols using secret keys require that the key is known only to the sender and the receiver of the encripted message. It is also required that the secret key can be efficiently replaced with a new one whenever considered necessary..

Distribution of the secret key is the critical phase in such communication because, if communication is eavesdropped, the key may become known to a third party.

Based on principles of quantum physics, protocols can be formulated which make it possible to detect eavesdropping.

Best known QKD protocols are:

- BB84 (Bennett and Brassard, 1984): protocol uses two complementary bases to prepare and measure photon polarisation states.
- E91 (Eckert, 1991): protocol makes use of entanglement of the polarization states of two photons (not covered in the present version of the course).

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The purpose of the protocol BB84 is to establish the secret encription key, i.e. a key known only to the sender (Alice, A) and the receiver (Bob, B).

The third party (Eve) is assumed to be able to eavesdrop the communication

Protocol BB84 allows A and B to detect potential eavesdropping and discard the established key if detection is positive.

Assumptions:

- Alice can send photons to Bob in any desired state of polarization.
- Bob can measure the polarization of incoming photons orienting the measurement apparatus in any way desired.
- Eve can intercept and measure the photons orienting the measurement apparatus in any way desired. Eve can also produce replacement photons in any state of polarisation.

Step zero: For the preparation and measurement of photon polarization states A and B agree to use the following bases:

- basis B_+ consisting of $|x\rangle$ i $|y\rangle$ (linear polarization oriented along x and y-axes) and
- basis $B_\times$ consisting of $|\pi/4\rangle$ and $|-\pi/4\rangle$ (linear polarization at angles $\pm 45^\circ$ with respect to x-axis).

They also agree on bit values assigned to the basis states:

Basis	Basis states	Bit value	Notation
B_+	$\|x\rangle$	0	$\|0_+\rangle$
	$\|y\rangle$	1	$\|1_+\rangle$
$B_\times$	$\|\pi/4\rangle = {}^1\!/\!\sqrt{2}\,(\|x\rangle + \|y\rangle)$	0	$\|0_\times\rangle$
	$\|-\pi/4\rangle = {}^1\!/\!\sqrt{2}\,(\|x\rangle - \|y\rangle)$	1	$\|1_\times\rangle$

Eve is assumed to be informed of this agreement.

key

Step one: Alice sends to Bob a sequence of photons such that the polarisation state is chosen from

- a random sequence of bit values 0 and 1
- and a random sequence of bases choices B_+ and $B_\times$.

This implies four different photon polarization states sent to Bob:

Alice:	Vrijednost bita	0		1	
	Baza	B_+	$B_\times$	B_+	$B_\times$
	Stanje polarizacije	$\|0_+\rangle$	$\|0_\times\rangle$	$\|1_+\rangle$	$\|1_\times\rangle$

Alice stores, but does not publish, the random sequences used.

Step two: Bob measures the polarization states chosing the basis according to his own random sequence (for now we assume no eavesdropping.)

Example: Communication without eavesdropping

Alice:	Bit value	...	0	1	0	1	1	...
	Basis	...	$B_\times$	$B_\times$	B_+	$B_\times$	B_+	...
	State	...	$\|0_\times\rangle$	$\|1_\times\rangle$	$\|0_+\rangle$	$\|1_\times\rangle$	$\|1_+\rangle$	...
Bob:	Basis	...	$B_\times$	B_+	B_+	B_+	B_+	...
	Measurement	...	0	${}^0\!/\!_1$	0	${}^0\!/\!_1$	1	...

If Bob uses the same base as Alice, measurement gives the bit value that Alice sent.

If bob uses the base that differs from the one Alice used, measurement gives the bit values 0 and 1 with equal probability (notation ${}^0\!/\!_1$).

Step three: Alice and Bob bob publish their random bases sequence (Alice does not publish her random bit sequence, nor does Bob publish his measurement results.) Then they independently compare the sequences and find the cases in which they used the same base. In these cases (cca 50%) they keep the bit values to form a key, while in the remaining cases they discard the bit values. The bit values that are not discarded comprise the secret key.

Example: Communication without eavesdropping with the established key

Alice:	$\cdots$	0	1	1	0	1	0	0	$\cdots$
	$\cdots$	B_+	B_+	$B_\times$	B_+	$B_\times$	$B_\times$	B_+	$\cdots$
	$\cdots$	$\|0_+\rangle$	$\|1_+\rangle$	$\|1_\times\rangle$	$\|0_+\rangle$	$\|1_\times\rangle$	$\|0_\times\rangle$	$\|0_+\rangle$	$\cdots$
Bob:	$\cdots$	B_+	$B_\times$	B_+	B_+	$B_\times$	B_+	B_+	$\cdots$
	$\cdots$	0	$0/1$	$0/1$	0	1	$0/1$	0	$\cdots$
Key:	$\cdots$	0	-	-	0	1	-	0	$\cdots$

Step four: Alice and Bob each publish a small part (a specimen) of the established key and compare one to another.

If the communication is eavesdropped, the keys disagree at approx. 25% bits. If the disagreement is found, Alice and Bob discard the key and look for the source of the problem.

Disagreement in bits occurs with 50% probability in cases where A and B used the same base, and Eve, randomly chosing the measurement base, happened to use the other one.

Primjer: Eavesdropped communication

Alice:	$\cdots$	0	1	1	1	0	$\cdots$
	$\cdots$	B_+	B_+	$B_\times$	$B_\times$	B_+	$\cdots$
	$\cdots$	$\|0_+\rangle$	$\|1_+\rangle$	$\|1_\times\rangle$	$\|1_\times\rangle$	$\|0_+\rangle$	$\cdots$
Eve:	$\cdots$	$B_\times$	B_+	B_+	$B_\times$	B_+	$\cdots$
	$\cdots$	$0/1$	1	$0/1$	1	0	$\cdots$
	$\cdots$	$\|0_\times\rangle/\|1_\times\rangle$	$\|1_+\rangle$	$\|0_+\rangle/\|1_+\rangle$	$\|1_\times\rangle$	$\|0_+\rangle$	$\cdots$
Bob:	$\cdots$	B_+	$B_\times$	B_+	$B_\times$	$B_\times$	$\cdots$
	$\cdots$	$0/1$	$0/1$	$0/1$	1	$0/1$	$\cdots$
Ključ:	$\cdots$	$0/1$	-	-	1	-	$\cdots$

Principles of Quantum Mechanics

Quantum Computers (SI)

October 24, 2025

PRINCIPLE 1: State vectors and state space

The state Φ of a physical system can be represented by a unit-normalized vector $|\Phi\rangle$ belonging to a N-dimensional complex (Hilbert) vector space $\mathcal{H}^{(N)}$.

The vector $|\Phi\rangle$ representing a state is called the *state vector*, and $\mathcal{H}^{(N)}$ is called the *state space*.

Dimension N of the state space reflects the complexity of the physical system being considered. Quantum information and computation is concerned with systems representable in finite-dimensional state spaces.

Example: Description of the state of a single quantum bit requires the state space of dimension $N = 2$. Description of the state of a n-qubit quantum computer requires $N = 2^n$.

Operations with vectors in complex vector space

- Inner product: $\langle\Psi|\Phi\rangle = \langle\Phi|\Psi\rangle^* \in \mathbb{C}$
- Orthogonality: $\langle\Phi|\Psi\rangle = 0$
- Normalization: $\langle\Phi|\Phi\rangle = 1$
- Orthonormal basis: $\{|n\rangle\,; n = 0, \ldots, N-1\}$, $\langle m|n\rangle = \delta_{mn}$
- Completeness:

$$|\Phi\rangle = \sum_{n=0}^{N-1} \lambda_n |n\rangle\,, \qquad \lambda_n = \langle n|\Phi\rangle \in \mathbb{C}$$

- If $|\Phi\rangle = \sum_n \lambda_n |n\rangle$, $|\Psi\rangle = \sum_n \mu |n\rangle$, then $\langle\Psi|\Phi\rangle = \sum_n \mu_n^* \lambda_n$
- Resolution of identity: $I = \sum_{n=0}^{N-1} |n\rangle\langle n|$

Measurement in quantum mechanics: Ψ-test

Ψ-test is a procedure consisting of the transition of the system prepared in an arbitrary state into the state $|\Psi\rangle$ followed by the detection of the system in the state $|\Psi\rangle$. The detection can be either positive or negative.

In the case of positive (negative) detection we say that the system has passed (has not passed) the $|\Psi\rangle$-test, or that it has (has not) been measured in the state $|\Psi\rangle$.

Example: If $|x\rangle$ represents the state of linear polarization of a photon oriented along the x-axis, then the x-test can be realized by a polarizer oriented along the x-axis followed by a detector.

If a photon is detected, we say it has passed the x-test. (However, having passed the x-test does not imply that before the measurement the state of the photon was $|x\rangle$.)

PRINCIPLE 2: Probability amplitude and probability

The *probability* that a system prepared in the state $|\Phi\rangle$ passes the Ψ-test, i.e. that it is meassured in the state $|\Psi\rangle$, is

$$p(\Phi \to \Psi) = \left|a(\Phi \to \Psi)\right|^2,$$

where

$$a(\Phi \to \Psi) = \langle\Psi|\Phi\rangle$$

is the *probability amplitude* for the transition of the system from $|\Phi\rangle$ to $|\Psi\rangle$.

Global phase

A complex number of modulus one is often called the *phase factor* and is written as $e^{i\varphi}$, where $\varphi \in \mathbb{R}$ is called the *phase*.

A phase factor multyplying a state vector is called the *global phase factor*.

A global phase factor multiplying a state vector $|\Phi\rangle$ has no effect on the probability that a system prepared in the state Φ passes a Ψ-test since, if $|\Phi'\rangle = e^{i\varphi}|\Phi\rangle$, then

$$\begin{aligned} p(\Phi' \to \Psi) &= |\langle\Psi|\Phi'\rangle|^2 \\ &= |e^{i\varphi}\langle\Psi|\Phi\rangle|^2 = |\langle\Psi|\Phi\rangle|^2 = p(\Phi' \to \Psi). \end{aligned}$$

Therefore state vectors that differ by a phase factor are understood to represent the same state of the system.

Projection of the state $|\Phi\rangle$ onto the state $|\Psi\rangle$

The state $|\Psi\rangle$ multiplied by the probability amplitude for the transition of the system from the state $|\Phi\rangle$ to the state $|\Psi\rangle$,

$$|\Psi\rangle\langle\Psi|\Phi\rangle,$$

is known as the *projection* of the state $|\Phi\rangle$ onto the state $|\Psi\rangle$.

The projection of a state onto another state is in general not normalized and therefore not a state vector.

Projection operator (projector)

Projection of an arbitrary state onto the state $|\Psi\rangle$ can be expressed using the *projection operator*

$$\mathcal{P}_\Psi = |\Psi\rangle\langle\Psi|.$$

Operating on the state $|\Phi\rangle$ it gives

$$|\mathcal{P}_\Psi\Phi\rangle = \mathcal{P}_\Psi|\Phi\rangle = (|\Psi\rangle\langle\Psi|)|\Phi\rangle = |\Psi\rangle\langle\Psi|\Phi\rangle.$$

Example: The photon polarization state can be given by the state-vector $|\Phi\rangle = \mu|x\rangle + \lambda|y\rangle$. Projection of the state $|\Phi\rangle$ onto the state $|x\rangle$ is:

$$|\mathcal{P}_x\Phi\rangle = \mathcal{P}_x|\Phi\rangle = |x\rangle\langle x|\left(\mu|x\rangle + \lambda|y\rangle\right) = \mu|x\rangle$$

Resolution of identity

Let $\{|n\rangle\,; n = 0, \ldots, N-1\}$ be an orthonormal basis of $\mathcal{H}^{(N)}$. The sum of projectors onto the basis states gives the identity operator,

$$I = \sum_{n=0}^{N-1} |n\rangle\langle n|.$$

The above relation is known as the *resolution of identity*.

It can be shown that resolution of identity is equivalent to the completeness relation, since

$$|\Phi\rangle = I|\Phi\rangle = \sum_{n=0}^{N-1} |n\rangle\langle n|\Phi\rangle = \sum_{n=0}^{N-1} \mu_n|n\rangle$$

confirming that $\mu_n = \langle n|\Phi\rangle$.

The commutator of operators A and B is the operator

$$[A, B] = AB - BA.$$

If $[A, B] = 0$ then A and B are said to commute. If $[A, B] \neq 0$ then A and B are said to not commute.

Example: Projectors onto orthogonal states commute.

Example: If $\langle x|y\rangle = 0$, projectors onto the states

$$|\theta\rangle = \cos\theta\,|x\rangle + \sin\theta\,|y\rangle \quad \text{and} \quad |\alpha\rangle = \cos\alpha\,|x\rangle + \sin\alpha\,|y\rangle$$

commute for $\theta - \alpha = k\pi/2$, $k \in \mathbb{Z}$.

Complementary bases: Bases $\{|n\rangle\,; n = 0, \ldots, N-1\}$ and $\{|\alpha\rangle\,; \alpha = 0, \ldots, N-1\}$ of $\mathcal{H}^{(N)}$ are said to be complementary if

$$|\,\langle\alpha|n\rangle\,|^2 = \frac{1}{N}.$$

Maximally incompatible tests: The tests corresponding to the basis states of any two complementary bases are said to be maximally incompatible.

Example: In the space of photon polarization states, the basis $\{|x\rangle\,, |y\rangle\}$ and the basis $\{|\theta\rangle\,, |\theta + \pi/2\rangle\}$ are complementary for $\theta = \pi/4$.

Hermitian operator

If in $\mathcal{H}^{(N)}$ a basis $\{|n\rangle\,; n = 0, \ldots, N-1\}$ can be found such that the operator $\hat{A}$ can be expressed as a linear combination of projectors $|n\rangle\langle n|$ with real coefficients A_n,

$$\hat{A} = \sum_{n=0}^{N-1} A_n\,|n\rangle\langle n|\,, \qquad A_n \in \mathbb{R},$$

then $\hat{A}$ is said to be a Hermitian operator.

The coefficients A_n are the eigenvalues of $\hat{A}$ and $|n\rangle$ are the eigenvectors.

Any operator with real eigenvalues is a Hermitian operator.

In quantum mechanics, measurable physical quantities are represented by Hermitian operators.

Compatible and incompatible tests

Let $\mathcal{P}_\Phi$ and $\mathcal{P}_\Psi$ be the projectors onto the states $|\Phi\rangle$ and $|\Psi\rangle$.

If $\mathcal{P}_\Phi$ and $\mathcal{P}_\Psi$ commute, i.e. if

$$[\mathcal{P}_\Phi, \mathcal{P}_\Psi] = 0,$$

then the Φ-test and the Ψ-test can be carried out simultaneously and these tests are said to be *compatible*.

If $\mathcal{P}_\Phi$ and $\mathcal{P}_\Psi$ do not commute, then the tests are *incompatible*.

Example: Tests that correspond to basis states are compatible.

Physical quantities

Let $A_n \in \mathbb{R}$ be the values of the physical quantity A corresponding to the basis states $\{|n\rangle\,; n = 0, \ldots, N-1\}$.

If the quantity A is measured in an ensemble of systems prepared in the state $|\Phi\rangle$, then the mean value of A is

$$\langle A\rangle = \sum_n A_n p(\Phi \to n) = \sum_n A_n |\,\langle n|\Phi\rangle\,|^2 = \sum_n A_n\,\langle\Phi|n\rangle\,\langle n|\Phi\rangle\,,$$

which can also be expressed as

$$\langle A\rangle = \langle\Phi|\hat{A}\Phi\rangle\,, \qquad \text{where} \qquad \hat{A} = \sum_n A_n\,|n\rangle\langle n|$$

is the operator respresenting the physical quantity A.

Expectation value of a Hermitian operator

Let $|\Phi\rangle$ be the state of the system and $\hat{M}$ the Hermitian operator. The quantity

$$\langle\hat{M}\rangle_\Phi = \langle\Phi|\hat{M}\Phi\rangle \in \mathbb{R}$$

is known as the expectation value of $\hat{M}$ in a system in the state $|\Phi\rangle$.

If $|\Phi\rangle = \sum_n \mu_n\,|n\rangle$ is the state of the system and $\hat{A} = \sum_n A_n\,|n\rangle\langle n|$, $A_n \in \mathbb{R}$, is a Hermitian operator representing the physical quantity $\hat{A}$, then

$$\langle\hat{A}\rangle_\Phi = \langle\Phi|\hat{A}\Phi\rangle = \cdots = \sum_n A_n p(\Phi \to n),$$

which is the mean value of the real quantity A that would be obtained in an ensemble of identical systems (or if it could be measured many times).

Example: A physical quantity related to photon polarization can be constructed by assigning the value $+1$ to the state $|x\rangle$ and assigning the value -1 to the state $|y\rangle$. The Hermitian operator representing this quantity then is

$$\hat{A} = (+1)\,|x\rangle\,\langle x| + (-1)\,|y\rangle\,\langle y|\,.$$

If the state vector representing the polarization state of the photon is

$$|R\rangle = \frac{1}{\sqrt{2}}(|x\rangle + \mathrm{i}\,|y\rangle),$$

then the expectation value of $\hat{A}$ is

$$\langle \hat{A} \rangle_R = \langle R|\hat{A}R\rangle = \cdots = 0.$$

PRINCIPLE 3: Unitarity of time evolution

Time-evolution of a quantum state,

$$|\Phi[t_1]\rangle \to |\Phi[t_2]\rangle\,,$$

is a *linear* transformation of the initial state that *preserves the norm* of the vector representing the state.

Linearity of the transformation implies that one can write

$$|\Phi[t_2]\rangle = U[t_2, t_1]\,|\Phi[t_1]\rangle\,,$$

where the *time-evolution operator* $U[t_2, t_1]$, since it preserves the norm of a vector, is said to be a *unitary operator*.

Some properties of the time-evolution unitary operator $U[t_2, t_1]$:

- Unitarity: if $|\Phi'\rangle = U\,|\Phi\rangle$, then $\langle\Phi'|\Phi'\rangle = \langle\Phi|\Phi\rangle$
- If $|\Phi'\rangle = U\,|\Phi\rangle$ and $|\Psi'\rangle = U\,|\Psi\rangle$, then $\langle\Psi'|\Phi'\rangle = \langle\Psi|\Phi\rangle$
- Invertibility: there exists "the inverse" U^{-1} such that $U^{-1}U = I$
- Composition of operators: $U[t_3, t_2]U[t_2, t_1] = U[t_3, t_1]$
- It also holds:

$$U^{-1}[t_2, t_1] = U[t_1, t_2], \qquad U[t, t] = I, \qquad \ldots$$

Schrödinger's equation (no derivation here)

From the requirement of unitarity of time evolution operator U, it follows that the differential equation

$$\mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{d}t}U[t, t_0] = H[t]U[t, t_0]$$

holds, where the operator H is Hermitian. Agreement with experiments is obtained if H represents the energy of the system and $\hbar = 1.05 \times 10^{-34}\,\mathrm{J\,s}$ is the reduced Planck's constant. The differential equation is known as *Schrödinger's equation*, and the energy operator H is known as the *Hamiltonian* of the system.

Schrödinger's equation can also be written as

$$\mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{d}t}\,|\Phi[t]\rangle = H[t]\,|\Phi[t]\rangle\,.$$

Example: Time-independent hamiltonian for the qubit

The Hamiltonian can be constructed as

$$H = \hbar\omega_0\,|0\rangle\,\langle 0| + \hbar\omega_1\,|1\rangle\,\langle 1|\,,$$

where the energies $\hbar\omega_0$ and $\hbar\omega_1$ correspond to the states $|0\rangle$ and $|1\rangle$ of the qubit. The general state of the qubit can be written as

$$|\Phi[t]\rangle = \mu[t]\,|0\rangle + \lambda[t]\,|1\rangle\,,$$

where at $t = 0$ we denote $\mu[0] = \mu_0$ and $\lambda[0] = \lambda_0$. Plugging H and $|\Phi[t]\rangle$ into the Schrödinger's equation we have

$$\mathrm{i}\hbar\left(\frac{\mathrm{d}}{\mathrm{d}t}\mu[t]\,|0\rangle + \frac{\mathrm{d}}{\mathrm{d}t}\lambda[t]\,|1\rangle\right) = \hbar\omega_0\mu[t]\,|0\rangle + \hbar\omega_1\lambda[t]\,|1\rangle\,.$$

It follows:

$$\frac{\mathrm{d}\mu}{\mu} = -\mathrm{i}\omega_0\mathrm{d}t, \qquad \frac{\mathrm{d}\lambda}{\lambda} = -\mathrm{i}\omega_1\mathrm{d}t$$

Integrating we get:

$$\mu[t] = \mu_0\mathrm{e}^{-\mathrm{i}\omega_0 t}, \qquad \lambda[t] = \lambda_0\mathrm{e}^{-\mathrm{i}\omega_1 t}.$$

The state of the qubit can now be written as

$$\begin{aligned}|\Phi[t]\rangle &= \mu_0\mathrm{e}^{-\mathrm{i}\omega_0 t}\,|0\rangle + \lambda_0\mathrm{e}^{-\mathrm{i}\omega_1 t}\,|1\rangle \\ &= \ldots = \mathrm{e}^{-\mathrm{i}\bar{\omega}t}\left(\mu_0\mathrm{e}^{-\mathrm{i}\delta t/2}\,|0\rangle + \lambda_0\mathrm{e}^{+\mathrm{i}\delta t/2}\,|1\rangle\right),\end{aligned}$$

where

$$\bar{\omega} = \frac{\omega_0 + \omega_1}{2}, \qquad \delta = \omega_0 - \omega_1.$$

Principles of Quantum Mechanics

Quantum Computers (SI)

17. listopada 2023.

easurements and probability amplitudes

Principle 1: Representation of states with vectors in Hilbert space

The state Φ of an isolated physical system can be represented by a unit-normalized vector $|\Phi\rangle$ belonging to a N-dimensional Hilbert space $\mathcal{H}^{(N)}$. The vector $|\Phi\rangle$ is referred to as the *state-vector*.

Interior product: $\langle\Psi|\Phi\rangle = \langle\Phi|\Psi\rangle^*$.

Orthogonality: $\langle\Phi|\Psi\rangle = 0$

Normalization: $\langle\Phi|\Phi\rangle = 1$.

Orthonormal basis: $\{|n\rangle\,; n = 0, \ldots, N-1\}$, $\langle m|n\rangle = \delta_{mn}$

Expansion: $|\Phi\rangle = \sum_{n=0}^{N-1} \mu_n |n\rangle$, $\mu \in \mathbb{C}$

Dimension N of the Hilbert space reflects the complexity of the physical system being considered.

Quantum information and computing is concerned with systems representable in finite-dimensional spaces.

Example: Dimensionality of Hilbert spaces in quantum computing

- Quantum mechanical description of a quantum bit requires the Hilbert space of dimension $N = 2$.
- Description of a n-qubit quantum computer requires the Hilbert space of dimension $N = 2^n$.

Ψ-test or measurement in the state $|\Psi\rangle$

Ψ-test is a procedure consisting of a transition of the system into the state $|\Psi\rangle$ and detection of the system in that state. The detection can be either positive or negative. In the case of positive (negative) detection we say that the system has passed (did not pass) the $|\Psi\rangle$-test, or that it has been measured in the state $|\Psi\rangle$.

- If the system is prepared in the state $|\Psi\rangle$ and is subjected to a Ψ-test, it is certain to pass the test (probability 1).
- If the system is prepared in a state orthogonal to $|\Psi\rangle$, it is certain to *not* pass the Ψ-test (probability 0).
- If the system is prepared in a general state Φ, the probability of passing a Ψ-test can be determined.

Example: If $|x\rangle$ represents the state of linear polarization of a photon oriented alng the x-axis, then the x-test can be realized by an analizer (polarizing filter) oriented along the x-axis followed by a detector.

If a photon is detected, we say it passed the x-test.

If it is not detected, we say it did not pass the x-test.

However, a photon passing the x-test does not imply that before the measurement the state of the photon was $|x\rangle$.

Principle 2: Probability amplitude and probability

The probability that a system in the state $|\Phi\rangle$ passes the Ψ-test, i.e. that it is meassured in the state $|\Psi\rangle$, is

$$p(\Phi \to \Psi) = |a(\Phi \to \Psi)|^2 = |\langle\Psi|\Phi\rangle|^2,$$

where

$$a(\Phi \to \Psi) = \langle\Psi|\Phi\rangle$$

is the *probability amplitude* for the transition of the system from the state $|\Phi\rangle$ to the state $|\Psi\rangle$.

n operators and Hermitian operators

Projection of the state $|\Phi\rangle$ onto the state $|\Psi\rangle$

The state $|\Psi\rangle$ multiplied by the probability amplitude for the transition of the system from the state $|\Phi\rangle$ to the state $|\Psi\rangle$,

$$|\Psi\rangle \langle\Psi|\Phi\rangle ,$$

is known as the projection of the state $|\Phi\rangle$ onto the state $|\Psi\rangle$.

The projection of a state onto another state is in general not a normalized vector and therefore does not represent a state of the system.

Projection operator (projector)

Projection of the state $|\Phi\rangle$ onto the state $|\Psi\rangle$ can be expressed as the result of the action of the *projection operator*

$$\mathcal{P}_\Psi = |\Psi\rangle \langle\Psi|$$

onto the state $|\Phi\rangle$,

$$|\mathcal{P}_\Psi \Phi\rangle = \mathcal{P}_\Psi |\Phi\rangle = (|\Psi\rangle \langle\Psi|) |\Phi\rangle = |\Psi\rangle \langle\Psi|\Phi\rangle .$$

Example: The photon polarization state can be given by the state-vector $|\Phi\rangle = \mu |x\rangle + \lambda |y\rangle$. Projection of the state $|\Phi\rangle$ onto the state $|x\rangle$ is:

$$|\mathcal{P}_x \Phi\rangle = \mathcal{P}_x |\Phi\rangle = |x\rangle \langle x| \left(\mu |x\rangle + \lambda |y\rangle\right) = \mu |x\rangle$$

Completeness relation

Let $\{|n\rangle ; n = 0, \ldots, N-1\}$ be an orthonormal basis of $\mathcal{H}^{(N)}$. The sum of projectors onto the basis states gives the identity operator,

$$\sum_{n=0}^{N-1} |n\rangle \langle n| = I.$$

The above relation is known as the *completeness relation*.

The completeness relation can be used to obtain the coefficients μ_n in the expansion of the vector $|\Phi\rangle$ in the basis $\{|n\rangle ; n = 1, \ldots, N\}$,

$$|\Phi\rangle = I |\Phi\rangle = \sum_{n=0}^{N-1} |n\rangle \langle n|\Phi\rangle = \sum_{n=0}^{N-1} \mu_n |n\rangle \quad \Longrightarrow \quad \mu_n = \langle n|\Phi\rangle .$$

The commutator (of two operators)

The *commutator* of operators A and B is the operator

$$[A, B] = AB - BA.$$

If the commutator is null the operators A and B are said to commute. If the commutator is non-null the operators A and B are said not to commute.

Example: Projectors onto the orthogonal states commute

Example: If $\langle x|y\rangle = 0$, projectors onto the states

$$|\theta\rangle = \cos\theta |x\rangle + \sin\theta |y\rangle \quad \text{and} \quad |\alpha\rangle = \cos\alpha |x\rangle + \sin\alpha |y\rangle$$

commute only for $\theta - \alpha = (k + 1/2)\pi$, $k \in \mathbb{Z}$.

Compatible and non-compatible tests

Let $\mathcal{P}_\Phi$ and $\mathcal{P}_\Psi$ be the projectors onto the states $|\Phi\rangle$ and $|\Psi\rangle$. If $\mathcal{P}_\Phi$ and $\mathcal{P}_\Psi$ commute, i.e. if

$$[\mathcal{P}_\Phi, \mathcal{P}_\Psi] = 0,$$

the Φ-test and the Ψ-test can be carried out simultaneously and these tests are said to be *compatible*. If $\mathcal{P}_\Phi$ and $\mathcal{P}_\Psi$ do not commute the tests are *incompatible*.

Example: Tests that correspond to basis states are compatible.

Maximally incompatible tests and complementary bases

Let $\{|n\rangle ; n = 0, \ldots, N-1\}$ and $\{|\alpha\rangle ; \alpha = 0, \ldots, N-1\}$ be the two different bases of $\mathcal{H}^{(N)}$. If

$$|\langle\alpha|n\rangle|^2 = \frac{1}{N},$$

the two bases are said to be complementary. The tests corresponding to the basis states of any two complementary bases are said to be maximally incompatible.

Example: In the space of photon polarization states, the basis $\{|x\rangle , |y\rangle\}$ and the basis $\{|\theta\rangle , |\theta + \pi/2\rangle\}$ are complementary for $\theta = \pi/4$.

Let $\{|n\rangle\,; n = 0, \ldots, N-1\}$ be the orthonormal basis of $\mathcal{H}^{(N)}$ chosen so that A_n are the values of a physical quantity A corresponding to basis states $|n\rangle$.

If in a system in the state $|\Phi\rangle$ the quantity A could be measured many times, the mean value of A would be

$$\sum_n A_n p(\Phi \to n) = \sum_n A_n |\langle n|\Phi\rangle|^2 = \sum_n A_n \langle\Phi|n\rangle \langle n|\Phi\rangle\,,$$

which can also be expressed as

$$\langle\Phi|\hat{A}\Phi\rangle\,, \qquad \text{where} \qquad \hat{A} = \sum_n A_n |n\rangle\langle n|$$

is the operator respresenting the physical quantity A.

Hermitian operator

If in $\mathcal{H}^{(N)}$ a basis $\{|n\rangle\,; n = 0, \ldots, N-1\}$ can be found such that the operator $\hat{A}$ can be expressed as a linear combination of projectors $|n\rangle\langle n|$ with real coefficients A_n,

$$A = \sum_{n=0}^{N-1} A_n |n\rangle\langle n|\,, \qquad A_n \in \mathbb{R},$$

then $\hat{A}$ is said to be a Hermitian operator. The coefficients A_n are the eigenvalues of $\hat{A}$ with $|n\rangle$ the eigenvectors.

Any operator with real eigenvalues is a Hermitian operator.

In quantum mechanics, measurable physical quantities are represented by Hermitian operators.

Expectation value of a Hermitian operator

Let $|\Phi\rangle$ be the state of the system and $\hat{M}$ the Hermitian operator. The quantity

$$\langle\hat{M}\rangle_\Phi = \langle\Phi|\hat{M}\Phi\rangle \in \mathbb{R}$$

is known as the expectation value of $\hat{M}$ in a system in the state $|\Phi\rangle$.

If $|\Phi\rangle = \sum_n \mu_n |n\rangle$ is the state of the system and $\hat{A} = \sum_n A_n |n\rangle\langle n|$, $A_n \in \mathbb{R}$, is a Hermitian operator representing the physical quantity $\hat{A}$, then

$$\langle\hat{A}\rangle_\Phi = \langle\Phi|\hat{A}\Phi\rangle = \cdots = \sum_n A_n p(\Phi \to n),$$

which is the mean value of the real quantity A that would be obtained if it could be measured many times.

Example: A physical quantity related to photon polarization can be constructed by assigning the value $+1$ to the state $|x\rangle$ and assigning the value -1 to the state $|y\rangle$. The corresponding Hermitian operator is

$$\hat{A} = (+1)\,|x\rangle\langle x| + (-1)\,|y\rangle\langle y|\,.$$

If the polarization state of a photon is

$$|R\rangle = \frac{1}{\sqrt{2}}(|x\rangle + \mathrm{i}\,|y\rangle),$$

the expectation value of $\hat{A}$ is

$$\langle\hat{A}\rangle_R = \langle R|\hat{A}R\rangle = \cdots = 0.$$

of states in time (Schrödinger's equation)

Principle 3: Unitarity of time evolution

Time-evolution of a quantum state,

$$|\Phi[t_1]\rangle \to |\Phi[t_2]\rangle\,,$$

is a *linear* transformation of the initial state that *preserves the norm* of the vector representing the state.

Linearity of the transformation implies that one can write

$$|\Phi[t_2]\rangle = U[t_2, t_1]\,|\Phi[t_1]\rangle\,,$$

where $U[t_2, t_1]$ is the *time-evolution operator* that, since it preserves the norm of a vector, is said to be a *unitary operator*.

Some properties of the time-evolution unitary operator $U[t_2, t_1]$:

- Unitarity: if $|\Phi'\rangle = U\,|\Phi\rangle$, then $\langle\Phi'|\Phi'\rangle = \langle\Phi|\Phi\rangle$
- If $|\Phi'\rangle = U\,|\Phi\rangle$ and $|\Psi'\rangle = U\,|\Psi\rangle$, then $\langle\Psi'|\Phi'\rangle = \langle\Psi|\Phi\rangle$
- Invertibility: there exists "the inverse" U^{-1} such that $U^{-1}U = I$
- Composition of operators: $U[t_3, t_2]U[t_2, t_1] = U[t_3, t_1]$
- It also holds:

$$U^{-1}[t_2, t_1] = U[t_1, t_2], \qquad U[t, t] = I, \qquad \ldots$$

Schrödinger's equation (no derivation here)

From the requirement of unitarity of thme evolution operator U, it follows that the differential equation

$$\mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{d}t}U[t,t_0] = \hat{H}[t]U[t,t_0]$$

holds, where the operator H is Hermitian. Agreement with experiments is obtained if H represents the energy of the system and $\hbar = 1.05 \times 10^{-34}\,\mathrm{J\,s}$ is the reduced Planck's constant. The differential equation is known as *Schrödinger's equation*, and the energy operator H is known as the *Hamiltonian* of the system.

Schrödinger's equation can also be written as

$$\mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{d}t}\left|\Phi[t]\right\rangle = \hat{H}[t]\left|\Phi[t]\right\rangle.$$

Example: Let energies $\hbar\omega_0$ and $\hbar\omega_1$ correspond to the states $|0\rangle$ and $|1\rangle$ of a qubit.

The Hamiltonian can be constructed as

$$\hat{H} = \hbar\omega_0\left|0\right\rangle\left\langle 0\right| + \hbar\omega_1\left|1\right\rangle\left\langle 1\right|.$$

The general state and the initial state at $t = 0$ can be written as

$$\left|\Phi[t]\right\rangle = \mu[t]\left|0\right\rangle + \lambda[t]\left|1\right\rangle, \qquad \mu[0] = \mu_0, \qquad \lambda[0] = \lambda_0.$$

Plugging $\hat{H}$ and $|\Phi[t]\rangle$ into the Schrödinger's equation we have

$$\mathrm{i}\hbar\left(\frac{\mathrm{d}}{\mathrm{d}t}\mu[t]\left|0\right\rangle + \frac{\mathrm{d}}{\mathrm{d}t}\lambda[t]\left|1\right\rangle\right) = \hbar\omega_0\mu[t]\left|0\right\rangle + \hbar\omega_1\lambda[t]\left|1\right\rangle.$$

It follows:

$$\frac{\mathrm{d}\mu}{\mu} = -\mathrm{i}\omega_0\mathrm{d}t, \qquad \frac{\mathrm{d}\lambda}{\lambda} = -\mathrm{i}\omega_1\mathrm{d}t$$

Integrating we get:

$$\mu[t] = \mu_0\mathrm{e}^{-\mathrm{i}\omega_0 t}, \qquad \lambda[t] = \lambda_0\mathrm{e}^{-\mathrm{i}\omega_1 t}.$$

The state of the qubit can now be written as:

$$\begin{aligned}\left|\Phi[t]\right\rangle &= \mu_0\mathrm{e}^{-\mathrm{i}\omega_0 t}\left|0\right\rangle + \lambda_0\mathrm{e}^{-\mathrm{i}\omega_1 t}\left|1\right\rangle \\ &= \mathrm{e}^{-\mathrm{i}\bar{\omega}t}\left(\mu_0\mathrm{e}^{-\mathrm{i}\delta t/2}\left|0\right\rangle + \lambda_0\mathrm{e}^{+\mathrm{i}\delta t/2}\left|1\right\rangle\right)\end{aligned}$$

gdje je

$$\bar{\omega} = \frac{\omega_0 + \omega_1}{2}, \qquad \delta = \omega_0 - \omega_1.$$

Matrix representation of states and operators

Quantum computers (SI)

October 24, 2023

dual vectors, and interior product ($N = 2$)

Orthonormal basis states $\{|0\rangle, |1\rangle\}$ in $\mathcal{H}^{(2)}$ are represented by single-column matrices (aslo called column-vectors)

$$\left|0\right\rangle = \begin{pmatrix}1\\0\end{pmatrix} \qquad \text{and} \qquad \left|1\right\rangle = \begin{pmatrix}0\\1\end{pmatrix}.$$

The general vector is then

$$\left|\Phi\right\rangle = \lambda\left|0\right\rangle + \mu\left|1\right\rangle = \begin{pmatrix}\lambda\\\mu\end{pmatrix}, \qquad \lambda,\mu\in\mathbb{C},$$

while its dual is represented by the single-row matrix (also called row-vector)

$$\left\langle\Phi\right| = \lambda^*\left\langle 0\right| + \mu^*\left\langle 1\right| = \begin{pmatrix}\lambda^* & \mu^*\end{pmatrix}.$$

The interior product of the vectors

$$\left|\Phi\right\rangle = \begin{pmatrix}\lambda\\\mu\end{pmatrix} \qquad \text{and} \qquad \left|\Psi\right\rangle = \begin{pmatrix}\nu\\\sigma\end{pmatrix}$$

is computed by *ordinary matrix multiplication* $(\cdot)$ of the matrix representing $|\Phi\rangle$ and the matrix representing $\langle\Psi|$,

$$\left\langle\Psi|\Phi\right\rangle = \begin{pmatrix}\nu^* & \sigma^*\end{pmatrix}\cdot\begin{pmatrix}\lambda\\\mu\end{pmatrix} = \nu^*\lambda + \sigma^*\mu.$$

Example: The projection of $|\Phi\rangle$ onto the state $|\Psi\rangle$:

$$\begin{aligned}\left|\Psi\right\rangle\left\langle\Psi|\Phi\right\rangle &= \begin{pmatrix}\nu\\\sigma\end{pmatrix}\left(\begin{pmatrix}\nu^* & \sigma^*\end{pmatrix}\cdot\begin{pmatrix}\lambda\\\mu\end{pmatrix}\right) = \begin{pmatrix}\nu\\\sigma\end{pmatrix}(\nu^*\lambda + \sigma^*\mu) \\ &= \begin{pmatrix}|\nu|^2\lambda + \nu\sigma^*\mu\\ \sigma\nu^*\lambda + |\sigma|^2\mu\end{pmatrix}\end{aligned}$$

projekcije ($N = 2$)

Let

$$|\Phi\rangle = \begin{pmatrix} \lambda \\ \mu \end{pmatrix} \qquad \text{and} \qquad |\Psi\rangle = \begin{pmatrix} \nu \\ \sigma \end{pmatrix}, \qquad \langle\Psi|\Psi\rangle = 1.$$

The projector onto $|\Psi\rangle$ is represented by the product of the matrix representing $\langle\Psi|$ and the matrix representing $|\Psi\rangle$,

$$\mathcal{P}_\Psi = |\Psi\rangle\langle\Psi| = \begin{pmatrix} \nu \\ \sigma \end{pmatrix} \cdot \begin{pmatrix} \nu^* & \sigma^* \end{pmatrix} = \begin{pmatrix} |\nu|^2 & \nu\sigma^* \\ \sigma\nu^* & |\sigma|^2 \end{pmatrix}$$

The projection of $|\Phi\rangle$ onto $|\Psi\rangle$ is then

$$\mathcal{P}_\Psi |\Phi\rangle = \begin{pmatrix} |\nu|^2 & \nu\sigma^* \\ \sigma\nu^* & |\sigma|^2 \end{pmatrix} \cdot \begin{pmatrix} \lambda \\ \mu \end{pmatrix} = \begin{pmatrix} |\nu|^2\lambda + \nu\sigma^*\mu \\ \sigma\nu^*\lambda + |\sigma|^2\mu \end{pmatrix}.$$

Example: The matrix representation of the projectors onto the states $|0\rangle$ i $|1\rangle$:

$$\mathcal{P}_0 = |0\rangle\langle 0| = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\mathcal{P}_1 = |1\rangle\langle 1| = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

The sum of the projectors gives the identity operator,

$$\mathcal{P}_0 + \mathcal{P}_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

(the completeness relation).

Example: State of right-handed circular polarization and the corresponding projector (using $|x\rangle = |0\rangle$, $|y\rangle = |1\rangle$):

$$|R\rangle = \frac{1}{\sqrt{2}}(|x\rangle + \mathrm{i}\,|y\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \mathrm{i} \end{pmatrix}$$

$$\mathcal{P}_R = |R\rangle\langle R| = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \mathrm{i} \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -\mathrm{i} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & -\mathrm{i} \\ \mathrm{i} & 1 \end{pmatrix}$$

State of left-handed circular polarization and the projector:

$$|L\rangle = \frac{1}{\sqrt{2}}(|x\rangle - \mathrm{i}\,|y\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -\mathrm{i} \end{pmatrix}$$

$$\mathcal{P}_L = |L\rangle\langle L| = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -\mathrm{i} \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & \mathrm{i} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & \mathrm{i} \\ -\mathrm{i} & 1 \end{pmatrix}$$

(Completeness: $\mathcal{P}_R + \mathcal{P}_L = I$)

operator i hermitsko konjugiranje

If $\{|n\rangle\,; n = 1, \ldots, N\}$ is the basis of $\mathcal{H}^{(N)}$, then state $|n\rangle$ is represented by the single-column matrix with 0 everywhere except in the n-th row where there is a 1, i.e. the j-th element from the top is

$$(|n\rangle)_j = \delta_{jn},$$

while the dual is represented by the single-row matrix whose i-th element from the left is $(\langle m|)_i = \delta_{mi}$.

If M is an operator in $\mathcal{H}^{(N)}$, and $(M)_{ij} = M_{ij}$ is the element in i-th row and j-th column of its matrix representation, we find

$$\langle m|\,M\,|n\rangle = \sum_{i=1}^{N}\sum_{j=1}^{N} (\langle m|)_i\,(M)_{ij}\,(|n\rangle)_j = \sum_{i=1}^{N}\sum_{j=1}^{N} \delta_{mi}\,M_{ij}\,\delta_{jn} = M_{mn}.$$

If Hermitian operator is constructed with

$$M = \sum_{n=1}^{N} a_n\,|n\rangle\langle n|\,, \qquad a_n \in \mathbb{R},$$

where the vectors $\{|n\rangle\,; n = 1, \ldots, N\}$ comprise an orthonormal basis, then in that particular basis the natrix representing M is a diagonal matrix,

$$M_{mn} = \langle m|\,M\,|n\rangle = a_m\delta_{mn}.$$

In the basis $\{|\alpha\rangle\,, \alpha = 1, \ldots, N\}$ whose vectors *do not* coincide with the eigenvectors of the Hermitian operator M, the elements of the matrix representing M have the property

$$M_{\alpha\beta} = \langle\alpha|\,M\,|\beta\rangle = \sum_{n=1}^{N} a_n\,\langle\alpha|n\rangle\,\langle n|\beta\rangle = \sum_{n=1}^{N} a_n\,\langle\beta|n\rangle^*\,\langle n|\alpha\rangle^*$$

$$= \Big(\sum_{n=1}^{N} a_n\,\langle\beta|n\rangle\,\langle n|\alpha\rangle\Big)^* = \langle\beta|\,M\,|\alpha\rangle^* = (M_{\beta\alpha})^*.$$

This means that the matrix representing a Hermitian operator is invariant with respect to transposition and complex conjugation (carried out in any order).

Hermitian conjugation of a matrix

Hermitian conjugation of a matrix is a transformation consisting of a transposition and complex conjugation of the matrix,

$$M^\dagger = (M^*)^T = (M^T)^*.$$

Hermitian conjugation of a matrix is denoted with the dagger symbol (†), while componentwise it holds

$$\left(M^\dagger\right)_{ij} = \left(M^*\right)_{ji} = \left((M)_{ji}\right)^*.$$

Matrix representation of a Hermitian operator

The matrix representation of a Hermitian operator M is invariant with respect to the operation of Hermitian conjugation,

$$M^\dagger = M \qquad \text{(Hermitian operator)}.$$

States and operators in $\mathcal{H}^{(2)}$

Quantum computers (SI)

October 24, 2023

ation of qubit states on the Bloch's sphere

Bloch's sphere

The general state of a qubit can be written in the form

$$|\Phi\rangle = \mathrm{e}^{-\mathrm{i}\varphi/2}\cos\frac{\vartheta}{2}\,|0\rangle + \mathrm{e}^{\mathrm{i}\varphi/2}\sin\frac{\vartheta}{2}\,|1\rangle$$

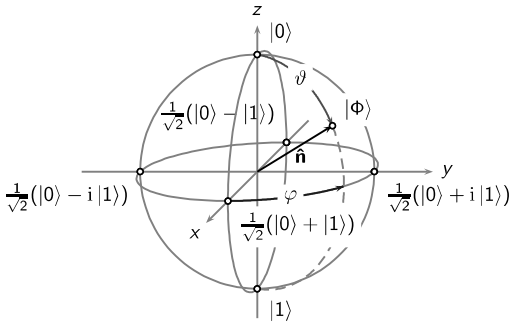
where $|0\rangle$ and $|1\rangle$ are the orthonormal basis vectors in $\mathcal{H}^{(2)}$, and

$$0 \le \varphi < 2\pi \qquad \text{i} \qquad 0 \le \vartheta \le \pi$$

can be interpreted as polar coordinates of so called *Bloch's sphere*.

The above form ensures normalization of the vector and also takes into account that $|\Phi\rangle$ and $\mathrm{e}^{\mathrm{i}\psi}|\Phi\rangle$ represent the same state.

State: $|\Phi\rangle = \mathrm{e}^{-\mathrm{i}\varphi/2}\cos\frac{\vartheta}{2}\,|0\rangle + \mathrm{e}^{\mathrm{i}\varphi/2}\sin\frac{\vartheta}{2}\,|1\rangle$



3D-vector: $\hat{\mathbf{n}} = \sin\vartheta\cos\varphi\,\hat{\mathbf{x}} + \sin\vartheta\sin\varphi\,\hat{\mathbf{y}} + \cos\vartheta\,\hat{\mathbf{z}}$

The point on the Bloch's sphere given by the 3D-vector $\hat{\mathbf{n}} = \sin\vartheta\cos\varphi\,\hat{\mathbf{x}} + \sin\vartheta\sin\varphi\,\hat{\mathbf{y}} + \cos\vartheta\,\hat{\mathbf{z}}$ corresponds to the state

$$|\Phi\rangle = \mathrm{e}^{-\mathrm{i}\varphi/2}\cos\frac{\vartheta}{2}\,|0\rangle + \mathrm{e}^{\mathrm{i}\varphi/2}\sin\frac{\vartheta}{2}\,|1\rangle\,.$$

3D-vector $-\hat{\mathbf{n}}$ which is of opposite orientation and is obtained by $\vartheta \to \pi - \vartheta$ and $\varphi \to \varphi \pm \pi$ corresponds to the state

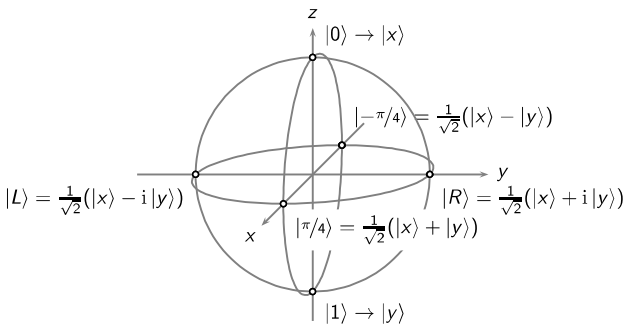
$$\begin{aligned}|\Phi_\perp\rangle &= \mathrm{e}^{-\mathrm{i}(\varphi\pm\pi)/2}\cos\frac{\pi-\vartheta}{2}\,|0\rangle + \mathrm{e}^{\mathrm{i}(\varphi\pm\pi)/2}\sin\frac{\pi-\vartheta}{2}\,|1\rangle \\ &= \mp\mathrm{i}\left(\mathrm{e}^{-\mathrm{i}\varphi/2}\sin\frac{\vartheta}{2}\,|0\rangle - \mathrm{e}^{\mathrm{i}\varphi/2}\cos\frac{\vartheta}{2}\,|1\rangle\right).\end{aligned}$$

Being orthogonal, $|\Phi\rangle$ i $|\Phi_\perp\rangle$ comprise a basis of $\mathcal{H}^{(2)}$.

Comments on qubits states representation on the Bloch's sphere:

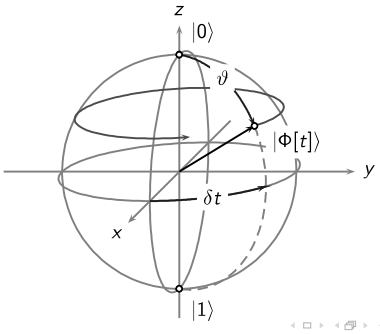
- Any pair of opposite points (poles) on the sphere can be taken as an orthonormal basis of the $\mathcal{H}^{(2)}$.
- Any choice of poles defines an equator; Tests corresponding to the states on the equator are maximally incompatible with the tests corresponding to the pole states.
- Any choice of two opposite points on the equator comprise an orthonormal basis complementary to the basis consisting of pole states.
- Simultaneously one can have only three complementary bases in $\mathcal{H}^{(2)}$ (these can be the bases defined by the pairs of points where x, y, and z-axis intersect the sphere.)

Example: Photon polarization states shown on BS:



Example: If the energies corresponding to the states $|0\rangle$ i $|1\rangle$ are $\hbar\omega_0$ and $\hbar\omega_1$,

$$|\Phi[t]\rangle = \mathrm{e}^{-\mathrm{i}\delta t/2}\cos\frac{\theta}{2}|0\rangle + \mathrm{e}^{\mathrm{i}\delta t/2}\sin\frac{\theta}{2}|1\rangle\,, \qquad \delta = \omega_0 - \omega_1.$$



ma) matrices

Projectors onto the states $|0\rangle = \binom{1}{0}$ and $|1\rangle = \binom{0}{1}$ (intersections of BS and z-axis) are

$$|0\rangle\langle 0| = \begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix} \qquad \text{and} \qquad |1\rangle\langle 1| = \begin{pmatrix}0 & 0\\ 0 & 1\end{pmatrix}.$$

These projectors are used to define the Hermitian operator

$$Z = (+1)\,|0\rangle\langle 0| \;+\; (-1)\,|1\rangle\langle 1| = \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}.$$

Eigenvalues of Z are $+1$ with eigenvector $|0\rangle$ and -1 with eigenvector $|1\rangle$.

Analogous procedure can be used to define the operators:

$$X = |+\rangle\langle +| - |-\rangle\langle -| = \cdots = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}$$

$$Y = |R\rangle\langle R| - |L\rangle\langle L| = \cdots = \begin{pmatrix}0 & -\mathrm{i}\\ \mathrm{i} & 0\end{pmatrix}$$

Action of X, Y, and Z, on a general state is equivalent to rotation of the state on the Bloch's sphere around the respective axis by π.

Alternative notation:

$$X = \sigma_1 = \sigma_x \qquad Y = \sigma_2 = \sigma_y \qquad Z = \sigma_3 = \sigma_z$$

Pauli or sigma-matrices

Matrices

$$\sigma_1 = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix}0 & -\mathrm{i}\\ \mathrm{i} & 0\end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix},$$

are known as *Pauli* or σ-*matrices*. They represent Hermitian operators in $\mathcal{H}^{(2)}$ whose eigenvectors coincide with the states of the three complementary bases in $\mathcal{H}^{(2)}$, and whose eigenvalues are ± 1.

Some properties:

$$\sigma_i^2 = I \qquad \sigma_1\sigma_2 = \mathrm{i}\sigma_3 \qquad \sigma_2\sigma_3 = \mathrm{i}\sigma_1 \qquad \sigma_3\sigma_1 = \mathrm{i}\sigma_2$$

Example: 3D-vectors $\pm\hat{\mathbf{n}}$ on the BS correspond to orthogonal states

$$|\Phi\rangle = \begin{pmatrix}\mathrm{e}^{-\mathrm{i}\varphi/2}\cos\frac{\vartheta}{2}\\ \mathrm{e}^{\mathrm{i}\varphi/2}\sin\frac{\vartheta}{2}\end{pmatrix} \qquad \text{and} \qquad |\Phi_\perp\rangle = \begin{pmatrix}\mathrm{e}^{-\mathrm{i}\varphi/2}\sin\frac{\vartheta}{2}\\ -\mathrm{e}^{\mathrm{i}\varphi/2}\cos\frac{\vartheta}{2}\end{pmatrix}$$

Define the Hermitian operator:

$$\sigma_{\hat{\mathbf{n}}} = (+1)\,|\Phi\rangle\langle\Phi| + (-1)\,|\Phi_\perp\rangle\langle\Phi_\perp| = \begin{pmatrix}\cos\vartheta & \mathrm{e}^{-\mathrm{i}\varphi}\sin\vartheta\\ \mathrm{e}^{\mathrm{i}\varphi}\sin\vartheta & -\cos\vartheta\end{pmatrix}$$

In the special cases $\hat{\mathbf{n}} = \hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$ one gets the Pauli matrices:

$$\begin{aligned}
&\hat{\mathbf{n}} = \hat{\mathbf{x}} && (\vartheta = \pi/2, \varphi = 0) && : && \sigma_{\hat{\mathbf{n}}} = X\\
&\hat{\mathbf{n}} = \hat{\mathbf{y}} && (\vartheta = \pi/2, \varphi = \pi/2) && : && \sigma_{\hat{\mathbf{n}}} = Y\\
&\hat{\mathbf{n}} = \hat{\mathbf{z}} && (\vartheta = 0) && : && \sigma_{\hat{\mathbf{n}}} = Z
\end{aligned}$$

Expansion of a Hermitian operator

Matrix representation of a general Hermitian operator M in $\mathcal{H}^{(2)}$ can be expanded in terms of Pauli matrices as

$$M = \lambda_0 I + \sum_{i=1}^{3} \lambda_i \sigma_i,$$

where I is the identity matrix, and λ_0, λ_1, λ_2, and λ_3, are real coefficients.

Example: Operator $\sigma_{\hat{\mathbf{n}}}$ can be expressed as

$$\sigma_{\hat{\mathbf{n}}} = \sin\vartheta\cos\varphi\,\sigma_1 + \sin\vartheta\sin\varphi\,\sigma_2 + \cos\vartheta\,\sigma_3,$$

where σ_1, σ_2, and σ_3, are Pauli matrices, and coefs $\lambda_1 = \sin\vartheta\cos\varphi$ etc. coincide with the components of the 3D-vector $\hat{\mathbf{n}}$.

as a realization of a quantum bit

Spin is a fundamental property of elementary particles.

Spin of a particle is quantized. The *spin quantum number* of a particle can have the values

$$s = 0, \frac{1}{2}, 1, \frac{3}{2}, \ldots,$$

while the *magnitude of the spin* is

$$S = \hbar\sqrt{s(s+1)},$$

where $\hbar = 1.05 \times 10^{-34}\,\mathrm{J\,s}$ is the reduced Planck's constant.

Electron, proton, and neutron, are $s = 1/2$ particles ($S = \frac{\sqrt{3}}{2}\hbar$).

Experimentally one can measure the projection of the spin onto the chosen spatial axis.

Projection of the spin of a particle with the spin quantum number s onto the chosen axis may have $2s+1$ different values. The possible values of the projection onto the z-axis are

$$S_z = -\hbar s, -\hbar(s-1), \ldots, \hbar(s-1), \hbar s.$$

If $s = 1/2$, the projection may have only two values,

$$S_z = \pm\frac{\hbar}{2}.$$

The projection of the spin of a "spin 1/2 particle" can be considered a realization of a quantum bit.

The projection of the spin of a "$s = 1/2$ particle" onto x, y, and z, axes is represented by Hermitian operators

$$S_x = \frac{\hbar}{2}\sigma_1, \qquad S_y = \frac{\hbar}{2}\sigma_2, \qquad S_z = \frac{\hbar}{2}\sigma_3,$$

while the vectorial nature of the spin can be represented by the operator

$$\mathbf{S} = \frac{\hbar}{2}\boldsymbol{\sigma}, \qquad \text{gdje je} \qquad \boldsymbol{\sigma} = \hat{\mathbf{x}}\,\sigma_1 + \hat{\mathbf{y}}\,\sigma_2 + \hat{\mathbf{z}}\,\sigma_3.$$

The expectation value of the spin vector in the system the state $|\Phi\rangle$ is

$$\langle\mathbf{S}\rangle = \langle\Phi|\mathbf{S}|\Phi\rangle = \cdots = \frac{\hbar}{2}\hat{\mathbf{n}},$$

where $\hat{\mathbf{n}}$ is a 3D-vector corresponding to the BS representation of the state $|\Phi\rangle$.

The Hermitian operator representing the square of the spin of a "$s = 1/2$ particle" is

$$\begin{aligned} S^2 = \mathbf{S}\cdot\mathbf{S} &= S_x^2 + S_y^2 + S_z^2 \\ &= \frac{\hbar^2}{4}\left(\sigma_1^2 + \sigma_2^2 + \sigma_3^2\right) \\ &= \frac{\hbar^2}{4}(I + I + I) \\ &= \frac{3\hbar^2}{4} I. \end{aligned}$$

We observe that the square of the spin does not depend on the spin state of the particle (we get $S = \frac{\sqrt{3}}{2}\hbar$ as expected, regardless of $\hat{\mathbf{n}}$).

Quantum entanglement

Quantum computers (SI)

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two qubits

The state of the isolated qubit A is given by the vector $|\phi_A\rangle$ in $\mathcal{H}_A^{(2)}$, where the basis $\{|0_A\rangle, |1_A\rangle\}$ can be used.

Isolated qubit B, state $|\phi_B\rangle$ in $\mathcal{H}_B^{(2)}$, basis $\{|0_B\rangle, |1_B\rangle\}$.

If A i B are in the states $|\phi_A\rangle$ and $|\phi_B\rangle$, the state of the system of two isolated qubits can be given by the *tensor product of the state vectors* $|\phi_A\rangle$ and $|\phi_B\rangle$ which is denoted

$$|\phi_B\rangle \otimes |\phi_A\rangle, \qquad \text{or more simply} \qquad |\phi_B \otimes \phi_A\rangle.$$

The above vector belongs to the *tensor product of Hilbert spaces* $\mathcal{H}_A$ and $\mathcal{H}_B$ which is a 4-dimensional Hilbert space and which is denoted with

$$\mathcal{H}_B \otimes \mathcal{H}_A.$$

Interior product of product states:

If: $|\phi\rangle = |\phi_B\rangle \otimes |\phi_A\rangle$ and $|\psi\rangle = |\psi_B\rangle \otimes |\psi_A\rangle$

then: $\langle\psi|\phi\rangle = \langle\psi_B|\phi_B\rangle \langle\psi_A|\phi_A\rangle$

In $\mathcal{H}_B \otimes \mathcal{H}_A$ one can construct an orthonormal basis $\{|ij\rangle = |i_B \otimes j_A\rangle; i,j = 0,1\}$, i.e.

$$\{ \quad |00\rangle = |0_B \otimes 0_A\rangle, \quad |01\rangle = |0_B \otimes 1_A\rangle, \\ |10\rangle = |1_B \otimes 0_A\rangle, \quad |11\rangle = |1_B \otimes 1_A\rangle \quad \}.$$

Orthonormality: $\langle kl|ij\rangle = \delta_{jl}\delta_{ik}$

A vector in $\mathcal{H}_B \otimes \mathcal{H}_A$ is then

$$|\phi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle,$$

where $\alpha, \beta, \gamma, \delta \in \mathbb{C}$. Normalization of $|\phi\rangle$ implies

$$|\alpha|^2 + |\beta|^2 + |\gamma|^2 + |\delta|^2 = 1.$$

If the states of the qubits A and B are

$$|\phi_A\rangle = \lambda_A|0_A\rangle + \mu_A|1_A\rangle \quad \text{and} \quad |\phi_B\rangle = \lambda_B|0_B\rangle + \mu_B|1_B\rangle,$$

the state of the two-qubit system is

$$\begin{aligned}|\phi\rangle = |\phi_B \otimes \phi_A\rangle &= \lambda_B\lambda_A|0_B \otimes 0_A\rangle + \lambda_B\mu_A|0_B \otimes 1_A\rangle \\ &\quad + \mu_B\lambda_A|1_B \otimes 0_A\rangle + \mu_B\mu_A|1_B \otimes 1_A\rangle \\ &= \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle,\end{aligned}$$

where $\alpha = \lambda_B\lambda_A$, $\beta = \lambda_B\mu_A$, $\gamma = \mu_B\lambda_A$ and $\delta = \mu_B\mu_A$.

As $|\phi_A\rangle$ and $|\phi_B\rangle$ are normalized, so is $|\phi\rangle$.

ed states

If the states of the qubits A and B are

$$|\phi_A\rangle = \lambda_A|0_A\rangle + \mu_A|1_A\rangle \quad \text{and} \quad |\phi_B\rangle = \lambda_B|0_B\rangle + \mu_B|1_B\rangle,$$

the state of the two-qubit system is

$$|\phi\rangle = |\phi_B \otimes \phi_A\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle,$$

where $\alpha = \lambda_B\lambda_A$, $\beta = \lambda_B\mu_A$, $\gamma = \mu_B\lambda_A$, and $\delta = \mu_B\mu_A$ and it holds

$$\alpha\delta = \beta\gamma,$$

which implies that the tensor product $|\phi_B \otimes \phi_A\rangle$ *is not* the general vector of $\mathcal{H}_B \otimes \mathcal{H}_A$.

Example: The Bell states

$$\begin{aligned}|\phi^+\rangle &= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \\ |\phi^-\rangle &= \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \\ |\psi^+\rangle &= \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \\ |\psi^-\rangle &= \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)\end{aligned}$$

can not be expressed as the tensor products of individual qubit state-vectors.

Product-states and entangled states of two or more qubits

- Multi-qubit states that can be represented by tensor products of individual qubit states are called *product-states*.
- Otherwise the state is said to be *entangled*.

Example: Two-qubit systems:

- Basis states $|00\rangle$, $|01\rangle$, $|10\rangle$ i $|11\rangle$ are product states by construction.
- Bell states are entangled.

Let M_A operate on states of A in $\mathcal{H}_A$ and M_B operate on states of B in $\mathcal{H}_B$.

The operator in $\mathcal{H}_B \otimes \mathcal{H}_A$ that corresponds to simultaneous operation of M_A and M_B is represented by the *tensor product of operators* denoted with

$$M_B \otimes M_A.$$

If the system is in a product state

$$|\phi_B \otimes \phi_A\rangle\,,$$

then

$$(M_B \otimes M_A)\,|\phi_B \otimes \phi_A\rangle = |M_B\phi_B \otimes M_A\phi_A\rangle\,.$$

Example: Let qubits A i B be realized by the projections of the spins of the two $s = 1/2$ particles onto the z-axis.

The projection of the spin of the particle B onto the z-axis in the two-particle system is represented by

$$\tfrac{\hbar}{2} Z \otimes I.$$

Acting on the basis states of $\mathcal{H}_B \otimes \mathcal{H}_A$ we have:

$$\begin{aligned}
(\tfrac{\hbar}{2} Z \otimes I)\,|00\rangle &= \tfrac{\hbar}{2}\,|Z0_B \otimes I0_A\rangle = \tfrac{\hbar}{2}\,|0_B \otimes 0_A\rangle = \tfrac{\hbar}{2}\,|00\rangle \\
(\tfrac{\hbar}{2} Z \otimes I)\,|01\rangle &= \tfrac{\hbar}{2}\,|Z0_B \otimes I1_A\rangle = \tfrac{\hbar}{2}\,|0_B \otimes 1_A\rangle = \tfrac{\hbar}{2}\,|01\rangle \\
(\tfrac{\hbar}{2} Z \otimes I)\,|10\rangle &= \tfrac{\hbar}{2}\,|Z1_B \otimes I0_A\rangle = -\tfrac{\hbar}{2}\,|1_B \otimes 0_A\rangle = -\tfrac{\hbar}{2}\,|10\rangle \\
(\tfrac{\hbar}{2} Z \otimes I)\,|11\rangle &= \tfrac{\hbar}{2}\,|Z1_B \otimes I1_A\rangle = -\tfrac{\hbar}{2}\,|1_B \otimes 1_A\rangle = -\tfrac{\hbar}{2}\,|11\rangle
\end{aligned}$$

Example: Let a two-qubit system be in the entangled state

$$|\psi^+\rangle = \frac{1}{\sqrt{2}}\left(|0_B \otimes 1_A\rangle + |1_B \otimes 0_A\rangle\right),$$

and let $M \otimes I$ be an operator. The expectation value is

$$\langle\psi^+|\,(M \otimes I)\,|\psi^+\rangle = \cdots = \frac{1}{2}\big(\,\langle 0_B|M0_B\rangle + \langle 1_B|M1_B\rangle\,\big).$$

It can be shown there is no single-qubit state-vector $|\psi_B\rangle = \lambda_B\,|0_B\rangle + \mu_B\,|1_B\rangle$ giving the same expectation value,

$$\begin{aligned}
\langle\psi_B|\,M\,|\psi_B\rangle = |\lambda_B|^2\,\langle 0_B|M0_B\rangle + |\mu_B|^2\,\langle 1_B|M1_B\rangle \\
+ \lambda_B^*\mu_B\,\langle 0_B|M1_B\rangle + \mu_B^*\lambda_B\,\langle 1_B|M0_B\rangle\,,
\end{aligned}$$

indicating the physical relevance of quantum entanglement.

Orthonormal basis vectors $\{|00\rangle\,,|01\rangle\,,|10\rangle\,,|11\rangle\}$ in $\mathcal{H}_B \otimes \mathcal{H}_A$ are represented by

$$|00\rangle = \begin{pmatrix}1\\0\\0\\0\end{pmatrix}, \quad |01\rangle = \begin{pmatrix}0\\1\\0\\0\end{pmatrix}, \quad |10\rangle = \begin{pmatrix}0\\0\\1\\0\end{pmatrix}, \quad |11\rangle = \begin{pmatrix}0\\0\\0\\1\end{pmatrix}.$$

The general vector $|\phi\rangle$ and the corresponding dual vector $\langle\phi|$ is

$$|\phi\rangle = \begin{pmatrix}\alpha\\\beta\\\gamma\\\delta\end{pmatrix} \qquad \text{i} \qquad \langle\phi| = \begin{pmatrix}\alpha^* & \beta^* & \gamma^* & \delta^*\end{pmatrix}.$$

If $(M_A)_{ij}$, $i,j = 0,1$, and $(M_B)_{pq}$, $p,q = 0,1$, are matrix elements of M_A i M_B, matrix elements of $M_B \otimes M_A$ are

$$(M_B \otimes M_A)_{pi;qj} = (M_B)_{pq}(M_A)_{ij}.$$

Example: If

$$M_A = \begin{pmatrix}\alpha & \beta\\ \gamma & \delta\end{pmatrix} \qquad \text{and} \qquad M_B = \begin{pmatrix}a & b\\ c & d\end{pmatrix},$$

then

$$M_B \otimes M_A = \begin{pmatrix}aM_A & bM_A\\ cM_A & dM_A\end{pmatrix} = \begin{pmatrix}a\alpha & a\beta & b\alpha & b\beta\\ a\gamma & a\delta & b\gamma & b\delta\\ c\alpha & c\beta & d\alpha & d\beta\\ c\gamma & c\delta & d\gamma & d\delta\end{pmatrix}$$

Example: Matrix representation of $Z \otimes Z$

$$Z \otimes Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Analogously:

$$X \otimes X = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \qquad Y \otimes Y = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}.$$

Example: If $\boldsymbol{\sigma} = X\,\hat{\mathbf{x}} + Y\,\hat{\mathbf{y}} + Z\,\hat{\mathbf{z}}$, then

$$M_{\text{swap}} = \frac{1}{2}(I + \boldsymbol{\sigma}_A \cdot \boldsymbol{\sigma}_B) = \frac{1}{2}(I + X \otimes X + Y \otimes Y + Z \otimes Z).$$

The matrix matrix representation of M_{swap} is

$$M_{\text{swap}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

The "swap behavior" of M_{swap}:

$$M_{\text{swap}}\,|ij\rangle = |ji\rangle$$

Example: The eigenvectors of

$$M_{\text{swap}} = \frac{1}{2}(I + \boldsymbol{\sigma}_A \cdot \boldsymbol{\sigma}_B)$$

are the vector

$$|\psi^-\rangle = \tfrac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

with the eigenvalue -1 and the *triplet* of vectors

$$|00\rangle, \qquad |\psi^+\rangle = \tfrac{1}{\sqrt{2}}(|01\rangle + |10\rangle), \qquad |11\rangle,$$

with the eigenvalue 1.

Example: Interaction energy resulting from relative orientation of the spins of two $s = 1/2$ particles can be given with the Hamiltonian

$$H = \hbar\omega\, M_{\text{swap}} = \frac{\hbar\omega}{2}(I + \boldsymbol{\sigma}_A \cdot \boldsymbol{\sigma}_B).$$

Eigenvectors of H are

$$|\psi^-\rangle = \tfrac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

with the eigenvalue $-\hbar\omega$ and the triplet

$$|00\rangle, \qquad |\psi^+\rangle = \tfrac{1}{\sqrt{2}}(|01\rangle + |10\rangle), \qquad |11\rangle,$$

with the eigenvalue $\hbar\omega$.

Example: Let the state of the two spins at $t = 0$ be

$$|\phi[0]\rangle = |01\rangle = \tfrac{1}{\sqrt{2}}(|\psi^+\rangle + |\psi^-\rangle),$$

which not an entangled state, but can be written as the superposition of two entangled states that are eigenvectors of the interaction Hamiltonian from an earlier slide. The time evolution of the state can be written as

$$|\phi[t]\rangle = \tfrac{1}{\sqrt{2}}\left(\mathrm{e}^{-\mathrm{i}\omega t}\,|\psi^+\rangle + \mathrm{e}^{\mathrm{i}\omega t}\,|\psi^-\rangle\right) = \cos\omega t\,|01\rangle - \mathrm{i}\sin\omega t\,|10\rangle.$$

We observe that at $\omega t = \pi/4$ the initially not entangled state becomes entangled

$$|\phi[\pi/4\omega]\rangle = \tfrac{1}{\sqrt{2}}\left(\,|01\rangle - \mathrm{i}\,|10\rangle\,\right).$$

Example: The sum of projections of the spins of two $s = 1/2$ particles onto the z-axis can be represented by the operator

$$S_z = \tfrac{\hbar}{2}(Z \otimes I + I \otimes Z).$$

The eigenvectors and eigenvalues of S_z are $|00\rangle$ with $\hbar$, $|11\rangle$ with $-\hbar$, and the dublet $\{|01\rangle, |10\rangle\}$, or $\{|\psi^+\rangle, |\psi^-\rangle\}$, or other linear combinations, with the eigenvalue 0.

Example: The eigenvectors of the total-spin-squared operator

$$S^2 = S_x^2 + S_y^2 + S_z^2 = \cdots = \hbar^2(I + M_{\text{swap}})$$

are $|\psi^-\rangle$ with the eigenvalue 0 and the vectors of the triplet $\{|00\rangle, |\psi^+\rangle, |11\rangle\}$ (or other linear combinations) with the eigenvalue $2\hbar^2$.

eorem

The no-cloning theorem: The unknown state of a quantum system can not be transfered from one system to another without compromising the state of the original system.

Colloquially: "there is no copy–paste of quantum information"

Proof: Assume cloning is possible, i.e. assume that there is a unitary operator U such that

$$U|\chi \otimes \phi\rangle = |\chi \otimes \chi\rangle$$

for arbitrary state $|\chi\rangle$. Then for arbitrary $|\chi_1\rangle$ and $|\chi_2\rangle$

$$U|\chi_1 \otimes \phi\rangle = |\chi_1 \otimes \chi_1\rangle\,,$$
$$U|\chi_2 \otimes \phi\rangle = |\chi_2 \otimes \chi_2\rangle\,.$$

Computing the interior product of left-hand-sides and the interior product of the right-hand-sides one gets

$$\langle\chi_1|\chi_2\rangle = \langle\chi_1|\chi_2\rangle^2$$

which does not hold in general, therefore we have contradiction.

Spin 1/2

Quantum computers (SI)

October 31, 2025

Sigma or Pauli matrices

Projectors onto the states $|0\rangle = \binom{1}{0}$ and $|1\rangle = \binom{0}{1}$ are

$$|0\rangle\langle 0| = \begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix} \qquad \text{and} \qquad |1\rangle\langle 1| = \begin{pmatrix}0 & 0\\ 0 & 1\end{pmatrix}.$$

These projectors are used to define the Hermitian operator

$$Z = (+1)\,|0\rangle\langle 0| \;+\; (-1)\,|1\rangle\langle 1| = \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}.$$

Eigenvalues of Z are $+1$ with eigenvector $|0\rangle$ and -1 with eigenvector $|1\rangle$.

Analogous procedure can be used to define the operators:

$$X = (+1)\,|+\rangle\langle +| + (-1)\,|-\rangle\langle -| = \cdots = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}$$

$$Y = (+1)\,|R\rangle\langle R| + (-1)\,|L\rangle\langle L| = \cdots = \begin{pmatrix}0 & -\mathrm{i}\\ \mathrm{i} & 0\end{pmatrix}$$

Operation of X, Y, and Z, on a general qubit state is equivalent to rotation of the state on the Bloch's sphere around the respective axis by π.

Alternative notation:

$$X = \sigma_x = \sigma_1 \qquad Y = \sigma_y = \sigma_2 \qquad Z = \sigma_z = \sigma_3$$

Pauli or sigma-matrices

Matrices

$$\sigma_1 = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix}0 & -\mathrm{i}\\ \mathrm{i} & 0\end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix},$$

are known as *Pauli* or σ*-matrices*. They represent Hermitian operators in $\mathcal{H}^{(2)}$ whose eigenvectors coincide with the states of the three complementary bases in $\mathcal{H}^{(2)}$, and whose eigenvalues are ± 1.

Some algebraic properties:

$$\sigma_j^2 = I \qquad \sigma_1\sigma_2 = \mathrm{i}\sigma_3 \qquad \sigma_2\sigma_3 = \mathrm{i}\sigma_1 \qquad \sigma_3\sigma_1 = \mathrm{i}\sigma_2$$

$$\sigma_j\sigma_k = \delta_{jk}I + \mathrm{i}\sum_{l=1}^{3}\varepsilon_{jkl}\sigma_l \qquad [\sigma_j, \sigma_k] = 2\mathrm{i}\sum_{l=1}^{3}\varepsilon_{jkl}\sigma_l$$

Expansion of a Hermitian operator

A Hermitian operator M in $\mathcal{H}^{(2)}$ can be expanded in terms of Pauli matrices as

$$M = \lambda_0 I + \sum_{i=1}^{3} \lambda_i \sigma_i,$$

where I is the identity matrix, and λ_0, λ_1, λ_2, and λ_3, are real coefficients.

Example: 3D-vectors $\pm\hat{\mathbf{n}}$ on the BS correspond to states:

$$|\Phi\rangle = \begin{pmatrix} e^{-i\varphi/2}\cos\frac{\vartheta}{2} \\ e^{i\varphi/2}\sin\frac{\vartheta}{2} \end{pmatrix} \qquad \text{and} \qquad |\Phi_\perp\rangle = \begin{pmatrix} e^{-i\varphi/2}\sin\frac{\vartheta}{2} \\ -e^{i\varphi/2}\cos\frac{\vartheta}{2} \end{pmatrix}$$

Define the Hermitian operator:

$$M = (+1)\,|\Phi\rangle\langle\Phi| + (-1)\,|\Phi_\perp\rangle\langle\Phi_\perp| = \begin{pmatrix} \cos\vartheta & e^{-i\varphi}\sin\vartheta \\ e^{i\varphi}\sin\vartheta & -\cos\vartheta \end{pmatrix}$$

In general:

$$M = \sin\vartheta\cos\varphi\, X + \sin\vartheta\sin\varphi\, Y + \cos\vartheta\, Z,$$

Special cases:

$$\begin{aligned} \hat{\mathbf{n}} = \hat{\mathbf{x}} \quad &(\text{or } \vartheta = \pi/2,\ \varphi = 0): \qquad M = X \\ \hat{\mathbf{n}} = \hat{\mathbf{y}} \quad &(\text{or } \vartheta = \pi/2,\ \varphi = \pi/2): \qquad M = Y \\ \hat{\mathbf{n}} = \hat{\mathbf{z}} \quad &(\text{or } \vartheta = 0): \qquad M = Z \end{aligned}$$

Spin 1/2 as a realization of a quantum bit

Spin is a fundamental property of elementary particles.

Spin of a particle is quantized. The **spin quantum number** of a particle can have the values

$$s = 0, \frac{1}{2}, 1, \frac{3}{2}, \ldots,$$

while the **magnitude of the spin** is

$$S = \hbar\sqrt{s(s+1)},$$

where $\hbar = 1.05 \times 10^{-34}\,\mathrm{J\,s}$ is the reduced Planck's constant.

Electron, proton, and neutron, are $s = 1/2$ particles ($S = \frac{\sqrt{3}}{2}\hbar$).

Experimentally one can measure the **projection of the spin onto the chosen spatial axis.**

Projection of the spin of a particle with the spin quantum number s onto the chosen axis may have $2s+1$ different values. The possible values of the projection onto the z-axis are

$$S_z = -\hbar s, -\hbar(s-1), \ldots, \hbar(s-1), \hbar s.$$

If $s = 1/2$, the projection may have only two values,

$$S_z = \pm\frac{\hbar}{2}.$$

The projection of the spin of a "spin 1/2 particle" can be considered a realization of a quantum bit.

The projection of the spin of a "$s = 1/2$ particle" onto x, y, and z, axes is represented by Hermitian operators

$$S_x = \frac{\hbar}{2}\sigma_1, \qquad S_y = \frac{\hbar}{2}\sigma_2, \qquad S_z = \frac{\hbar}{2}\sigma_3,$$

while the vectorial nature of the spin can be represented by the operator

$$\mathbf{S} = \frac{\hbar}{2}\boldsymbol{\sigma}, \qquad \text{where} \qquad \boldsymbol{\sigma} = \hat{\mathbf{x}}\,\sigma_1 + \hat{\mathbf{y}}\,\sigma_2 + \hat{\mathbf{z}}\,\sigma_3.$$

The expectation value of the spin vector in the system prepared in the state $|\Phi\rangle$ is

$$\langle\mathbf{S}\rangle = \langle\Phi|\mathbf{S}|\Phi\rangle = \cdots = \frac{\hbar}{2}\,\hat{\mathbf{n}},$$

where $\hat{\mathbf{n}}$ is a 3D-vector corresponding to $|\Phi\rangle$.

The Hermitian operator representing the square of the spin of a "$s = 1/2$ particle" is

$$\begin{aligned} S^2 = \mathbf{S}\cdot\mathbf{S} &= S_x^2 + S_y^2 + S_z^2 \\ &= \frac{\hbar^2}{4}\left(\sigma_1^2 + \sigma_2^2 + \sigma_3^2\right) \\ &= \frac{\hbar^2}{4}\left(I + I + I\right) \\ &= \frac{3\hbar^2}{4} I. \end{aligned}$$

We observe that the square of the spin does not depend on the spin state of the particle (we get $S = \frac{\sqrt{3}}{2}\hbar$ as expected, regardless of $\hat{\mathbf{n}}$).

Mixed states and entanglement

Quantum computers (SI)

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In an ensemble of systems prepared so that each system is in the state $|\phi\rangle$, the mean value of the physical quantity A is given by

$$\bar{A} = \langle \hat{A} \rangle_\phi = \langle \phi | \hat{A} | \phi \rangle \,,$$

where $\hat{A}$ is the Hermitian operator representing A.

In an ensemble of systems prepared so that an individual system has the probability P_i of being prepared in the state $|\phi_i\rangle$, the mean of A is

$$\bar{A} = \sum_i P_i \langle \phi_i | \hat{A} | \phi_i \rangle \,,$$

where it is assumed that $\sum_i P_i = 1$, and the states $|\phi_i\rangle$ need not be orthogonal.

Pure state of a system

System whose state can be given by a state vector is said to be in a *pure state*.

Mixed state of a system

System whose state is known only up to the probability P_i of being in the state $|\phi_i\rangle$ is said to be in a *mixed state*.

State operator (or density operator, or density matrix)

The state of a system can be specified by the operator

$$\hat{\rho} = \sum_i P_i \, |\phi_i\rangle \, \langle \phi_i|$$

where P_i are the probabilities of the system being in the state $|\phi_i\rangle$.

Example: The state of a system in a pure state that is represented by the state-vector $|\phi\rangle$ can be represented by the state operator

$$\hat{\rho} = |\phi\rangle \, \langle \phi| \,.$$

Example: Non-uniqueness of the interpretation of mixed states

The state operator

$$\hat{\rho} = \frac{1}{2} |0\rangle \, \langle 0| + \frac{1}{2} |1\rangle \, \langle 1|$$

that can be interpreted as representing the mixed state of $|0\rangle$ and $|1\rangle$ with probabilities $1/2$ can also be written as

$$\begin{aligned} \frac{1}{3} |0\rangle \, \langle 0| + \frac{1}{3} \frac{(|0\rangle + \sqrt{3} \, |1\rangle)}{2} \frac{(\langle 0| + \sqrt{3} \, \langle 1|)}{2} \\ + \frac{1}{3} \frac{(|0\rangle - \sqrt{3} \, |1\rangle)}{2} \frac{(\langle 0| - \sqrt{3} \, \langle 1|)}{2} \end{aligned}$$

which can be interpreted as the mixture of three non-orthogonal states with probabilities $1/3$.

The probability of passing a test

If the state of the system is given by the state operator $\hat{\rho}$, the probability of the system being measured in the state represented by the state vector $|\psi\rangle$ is

$$p = \langle \psi | \hat{\rho} | \psi \rangle \,.$$

Proof: Using $\hat{\rho} = \sum_i P_i \, |\phi_i\rangle \, \langle \phi_i|$ and Born's rule,

$$\begin{aligned} p = \sum_i P_i |\langle \psi | \phi_i \rangle|^2 &= \sum_i P_i \, \langle \psi | \phi_i \rangle \, \langle \phi_i | \psi \rangle \\ &= \langle \psi | \left(\sum_i P_i \, |\phi_i\rangle \, \langle \phi_i| \right) |\psi\rangle \\ &= \langle \psi | \, \hat{\rho} \, |\psi\rangle \end{aligned}$$

The mean of A

If the state of the system is given by the state operator $\hat{\rho}$, the mean value of a quantity represented by $\hat{A}$ is given by

$$\bar{A} = \langle \hat{A} \rangle = \mathrm{Tr}(\hat{\rho}\hat{A}).$$

Proof: Compute the trace using the base $\{|n\rangle\}$,

$$\begin{aligned}\mathrm{Tr}(\hat{\rho}\hat{A}) = \sum_n \langle n| \hat{\rho}\hat{A} |n\rangle &= \sum_n \langle n| \left(\sum_i P_i |\phi_i\rangle \langle \phi_i| \right) \hat{A} |n\rangle \\ &= \sum_i P_i \langle \phi_i| \left(\hat{A} \sum_n |n\rangle \langle n| \right) |\phi_i\rangle \\ &= \sum_i P_i \langle \phi_i| \hat{A} |\phi_i\rangle\end{aligned}$$

Let a compound system consist of the subsystems A and B. Let $\{|m\rangle\}$ (Latin indices) be the orthonormal basis of A, and let $\{|\mu\rangle\}$ (Greek indices) be the orthonormal basis of B.

Using $\{|\mu m\rangle = |\mu\rangle \otimes |m\rangle\}$ as the basis of the compound system, its pure state can be given by the state vector

$$|\Phi\rangle = \sum_{\mu,m} \alpha_{\mu m} |\mu m\rangle ,$$

where $\alpha_{\mu m} = \langle \mu m | \Phi \rangle$, or by the state operator

$$\hat{\rho} = |\Phi\rangle \langle \Phi| = \sum_{\mu,m} \sum_{\nu,n} \alpha_{\mu m} \alpha^*_{\nu n} |\mu m\rangle \langle \nu n| ,$$

whose matrix element is

$$(\hat{\rho})_{\mu m\, \nu n} = \langle \mu m | \hat{\rho} | \nu n \rangle = \langle \mu m | \Phi \rangle \langle \Phi | \nu n \rangle = \alpha_{\mu m} \alpha^*_{\nu n}.$$

In the compound system, the expectation value of the operator $\hat{M}_A$ operating on the states of the subsystem A is

$$\begin{aligned}\langle \Phi| (\hat{I}_B \otimes \hat{M}_A) |\Phi\rangle &= \sum_{\mu,m} \sum_{\nu,n} \alpha_{\nu n} \alpha^*_{\mu m} \langle \mu m| (\hat{I}_B \otimes \hat{M}_A) |\nu n\rangle \\ &= \sum_{n,m} \sum_{\mu,\nu} (\hat{\rho})_{\nu n\, \mu m} \ \langle \mu|\nu\rangle \langle m|\hat{M}_A|n\rangle \\ &= \sum_{n,m} \sum_{\mu,\nu} (\hat{\rho})_{\nu n\, \mu m} \ \delta_{\mu\nu} \ (\hat{M}_A)_{mn} \\ &= \sum_{n,m} \left(\sum_{\mu} (\hat{\rho})_{\mu n\, \mu m} \right) (\hat{M}_A)_{mn} \\ &= \sum_{n,m} (\hat{\rho}_A)_{nm} (\hat{M}_A)_{mn} \\ &= \mathrm{Tr}[\hat{\rho}_A \hat{M}_A].\end{aligned}$$

The operator $\hat{\rho}_A$ operating on the states of the subsystem A and whose matrix elements were found to be

$$(\hat{\rho}_A)_{nm} = \langle n| \hat{\rho}_A |m\rangle = \sum_{\mu} (\hat{\rho})_{\mu n\, \mu m}$$

is known as the *reduced state operator of the subsystem A*. It is obtained by "partial tracing" of the state operator of the compuound system over the index of the subsystem B (Greek index).

Analogously, the expectation value of $\hat{M}_B$ operating on the states of the subsystem B can be given by

$$\langle \Phi| (\hat{M}_B \otimes \hat{I}_A) |\Phi\rangle = \mathrm{Tr}[\hat{\rho}_B \hat{I}_A],$$

where the matrix elements of the reduced state operator of B are

$$(\hat{\rho}_B)_{\nu\mu} = \langle \nu| \hat{\rho}_B |\mu\rangle = \sum_{m} (\hat{\rho})_{\nu m\, \mu m}.$$

Example: If the system of qubits A and B is in the state

$$|\Phi\rangle = \alpha_{00} |00\rangle + \alpha_{01} |01\rangle + \alpha_{10} |10\rangle + \alpha_{11} |11\rangle ,$$

the reduced state operator of qubit A is

$$\hat{\rho}_A = \begin{pmatrix} |\alpha_{00}|^2 + |\alpha_{10}|^2 & \alpha_{00}\alpha^*_{01} + \alpha_{10}\alpha^*_{11} \\ \alpha^*_{00}\alpha_{01} + \alpha^*_{10}\alpha_{11} & |\alpha_{01}|^2 + |\alpha_{11}|^2 \end{pmatrix},$$

while the reduced state operator of qubit B is

$$\hat{\rho}_B = \begin{pmatrix} |\alpha_{00}|^2 + |\alpha_{01}|^2 & \alpha_{00}\alpha^*_{10} + \alpha_{01}\alpha^*_{11} \\ \alpha^*_{00}\alpha_{10} + \alpha^*_{01}\alpha_{11} & |\alpha_{10}|^2 + |\alpha_{11}|^2 \end{pmatrix}.$$

Example: If qubits A and B are in the states

$$|\phi_A\rangle = \lambda_A |0_A\rangle + \mu_A |1_A\rangle \quad \text{and} \quad |\phi_B\rangle = \lambda_B |0_B\rangle + \mu_B |1_B\rangle ,$$

then the compund system BA is in the product state

$$|\phi_B \otimes \phi_A\rangle = \alpha_{00} |00\rangle + \alpha_{01} |01\rangle + \alpha_{10} |10\rangle + \alpha_{11} |11\rangle ,$$

where $\alpha_{00} = \lambda_B\lambda_A$, $\alpha_{01} = \lambda_B\mu_A$, $\alpha_{10} = \mu_B\lambda_A$ i $\alpha_{11} = \mu_B\mu_A$.

The reduced state operators of A and B are

$$\rho_A = \begin{pmatrix} |\lambda_A|^2 & \lambda_A\mu^*_A \\ \lambda^*_A\mu_A & |\mu_A|^2 \end{pmatrix} \quad \text{and} \quad \rho_B = \begin{pmatrix} |\lambda_B|^2 & \lambda_B\mu^*_B \\ \lambda^*_B\mu_B & |\mu_B|^2 \end{pmatrix}$$

Classical and Quantum Information

Quantum computers (SI)

January 20, 2025

1. Single systems
   - Classical systems: Probabilistic states and operati
   - Quantum systems: Unitary operations
   - Single qubit states and operators
2. Multiple systems
   - Classical systems
   - Quantum systems
3. Logical circuits
   - Reversible formulation of classical logic
   - Quantum logical circuits
4. Entanglement

A classical carrier of information is a physical system that can be in one of a finite number of its *classical states*.

Example: Let X be the classical information carrying system and let Σ be the corresponding set of classical states.

- If X is a bit, then $\Sigma = \{0, 1\}$ (binary alphabet)
- If X is a fan switch, then $\Sigma = \{\text{off}, \text{lo}, \text{mid}, \text{hi}\}$
- If X is a six-sided die, then $\Sigma = \{1, 2, 3, 4, 5, 6\}$
- If X is a byte, then $\Sigma = \{0, \ldots, FF\}$
- ...

Classical bit states 0 and 1 are represented by *state vectors*

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad \text{and} \qquad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

If our knowledge or belief about the state of a bit is uncertain, i.e. if it is given in terms of probabilities $p(0)$ and $p(1)$, the state can be specified by the *probability vector*

$$p(0)\,|0\rangle + p(1)\,|1\rangle = \begin{pmatrix} p(0) \\ p(1) \end{pmatrix}.$$

Such state is said to be a *probabilistic state*. It is understood that $p(0) + p(1) = 1$. Generalization to systems with more than two classical states is obvious.

Measurement of a classical system yields one of its classical states.

If before the measurement the state of the system is probabilistic, the measurement turns it into the classical state $|a\rangle \in \Sigma$ obtained in the measurement.

Example: Flipping of a fair coin

- State before looking at a flipped coin (belief):

$$\frac{1}{2}\,|\text{heads}\rangle + \frac{1}{2}\,|\text{tails}\rangle$$

- State after examining the flipped coin (measurement):

$$|\text{heads}\rangle \qquad \text{or} \qquad |\text{tails}\rangle$$

Deterministic operations turn classical states into classical states.

Example: All deterministic operations on a classical bit state

$$M_1 = |0\rangle\langle 0| + |0\rangle\langle 1| = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \qquad \text{(constant, 0)}$$

$$M_2 = |0\rangle\langle 0| + |1\rangle\langle 1| = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \qquad \text{(balanced, identity, } I\text{)}$$

$$M_3 = |1\rangle\langle 0| + |0\rangle\langle 1| = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \qquad \text{(balanced, NOT, } X\text{)}$$

$$M_4 = |1\rangle\langle 0| + |1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} \qquad \text{(constant, 1)}$$

Matrix representations of deterministic operations have exactly one 1 in each column and 0 elsewhere.

Probabilistic operations turn classical states into probabilistic states. Matrices representing probabilistic operations are known as *stochastic matrices*.

Example: The probabilistic operation on a classical bit can be specified with

$$|0\rangle \longrightarrow p_{00}\,|0\rangle + p_{01}\,|1\rangle \quad \text{and} \quad |1\rangle \longrightarrow p_{10}\,|0\rangle + p_{11}\,|1\rangle\,.$$

The corresponding stochastic matrix is

$$\begin{aligned} M &= \left(p_{00}\,|0\rangle + p_{01}\,|1\rangle\right)\langle 0| + \left(p_{10}\,|0\rangle + p_{11}\,|1\rangle\right)\langle 1| \\ &= \begin{pmatrix} p_{00} & p_{10} \\ p_{01} & p_{11} \end{pmatrix}. \end{aligned}$$

Some general properties of probabilistic operations and stochastic matrices:

- Probabilistic operation turns a probability vector into another probability vector (easy to prove).
- All entries in a stochastic matrix are nonnegative real numbers.
- Sum of entries in each column of the matrix equals 1.
- Each column of the matrix is a probability vector.

With every quantum information carrying system a set of classical states, $\Sigma = \{|i\rangle : i = 0, \ldots, N-1\}$, can be associated so that the *quantum state* $|\psi\rangle$ of the system can be given by

$$|\psi\rangle = \alpha_0\,|0\rangle + \alpha_1\,|1\rangle + \cdots + \alpha_{N-1}\,|N-1\rangle = \begin{pmatrix} \alpha_0 \\ \vdots \\ \alpha_{N-1} \end{pmatrix},$$

where

- classical states are assumed orthonormal, $\langle i|j\rangle = \delta_{ij}$,
- coefficients α_i are complex numbers, $\alpha_i \in \mathbb{C}$,
- the state $|\psi\rangle$ is normalized, $\langle\psi|\psi\rangle = 1$, which implies $|\alpha_0|^2 + |\alpha_1|^2 + \cdots + |\alpha_{N-1}|^2 = 1$.

In the so called *standard basis measurement* of a quantum system, the result of the measurement is one of its classical states.

If the quantum state of the system is $|\psi\rangle = \sum_i \alpha_i\,|i\rangle$, then the probability of measuring the classical state $|i\rangle$ is $|\alpha_i|^2$. This result is known as the *Born rule*.

Example: The classical states of a quantum bit are $|0\rangle$ and $|1\rangle$, so the quantum state is

$$|\psi\rangle = \lambda\,|0\rangle + \mu\,|1\rangle = \begin{pmatrix} \lambda \\ \mu \end{pmatrix}, \qquad \lambda, \mu \in \mathbb{C}, \qquad |\lambda|^2 + |\mu|^2 = 1.$$

The probability of measuring the state $|0\rangle$ is $|\lambda|^2$, for $|1\rangle$ it is $|\mu|^2$,

Operations on quantum states are restricted to so called *unitary operations* that preserve the normalization of the states. Unitary operators, and corresponding matrix representations known as unitary matrices, have the properties:

- $U^\dagger U = I = UU^\dagger$
- Unitary operators are invertible, $U^{-1} = U^\dagger$.
- Unitary operators are normal, which implies they have spectral decomposition.
- Eigenvalues are complex numbers with modulus 1.

Single qubit states

- Computational basis: $\Sigma = \{|0\rangle, |1\rangle\}$
- Plus and minus state:

$$|+\rangle = \frac{1}{\sqrt{2}}\,|0\rangle + \frac{1}{\sqrt{2}}\,|1\rangle \qquad |-\rangle = \frac{1}{\sqrt{2}}\,|0\rangle - \frac{1}{\sqrt{2}}\,|1\rangle$$

- Right and left state:

$$|R\rangle = \frac{1}{\sqrt{2}}\,|0\rangle + \frac{\mathrm{i}}{\sqrt{2}}\,|1\rangle \qquad |L\rangle = \frac{1}{\sqrt{2}}\,|0\rangle - \frac{\mathrm{i}}{\sqrt{2}}\,|1\rangle$$

- State with Bloch sphere coordinates ϑ and φ:

$$|\Psi\rangle = \cos\frac{\vartheta}{2}|0\rangle + e^{i\varphi}\sin\frac{\vartheta}{2}|1\rangle$$

- State orthogonal to $|\Psi\rangle$:

$$|\Psi_\perp\rangle = \sin\frac{\vartheta}{2}|0\rangle - e^{i\varphi}\cos\frac{\vartheta}{2}|1\rangle$$

- A pair of orthogonal states:

$$|\chi^\pm\rangle = \cos\left[\frac{\pi}{4} \pm \frac{\gamma}{2}\right]|0\rangle \pm e^{i\varphi}\sin\left[\frac{\pi}{4} \pm \frac{\gamma}{2}\right]|1\rangle$$

Single qubit operators

- Identity operator:

$$I = |0\rangle\langle 0| + |1\rangle\langle 1| = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

Properties: unitary, Hermitian, eigenvalue 1 with all vectors, classical deterministic operation identity

- Pauli X:

$$X = |+\rangle\langle +| - |-\rangle\langle -| = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Properties: unitary, Hermitian, eigenvalues ± 1 with eigenvectors $|\pm\rangle$, classical deterministic operation NOT

- Pauli Y:

$$Y = |R\rangle\langle R| - |L\rangle\langle L| = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

Properties: unitary, Hermitian, eigenvalue $+1$ with $|R\rangle$, -1 with $|L\rangle$, not a classical operation

- Pauli Z:

$$Z = |0\rangle\langle 0| - |1\rangle\langle 1| = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Properties: unitary, Hermitian, eigenvalue $+1$ with $|0\rangle$, -1 with $|1\rangle$, $Z^2 = I$, $Z|\pm\rangle = |\mp\rangle$, $Z|R\rangle = |L\rangle$, $Z|L\rangle = |R\rangle$, not a classical operation

- Hadamard operator H:

denote: $|\chi^\pm\rangle = \cos\left[\frac{\pi}{4} \pm \frac{\pi}{8}\right]|0\rangle \pm \sin\left[\frac{\pi}{4} \pm \frac{\pi}{8}\right]|1\rangle$

$$H = |\chi^+\rangle\langle\chi^+| - |\chi^-\rangle\langle\chi^-| = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

Properties: unitary, Hermitian, eigenvalues ± 1 with $|\chi^\pm\rangle$, $H^2 = I$, $H|0\rangle = |+\rangle$, $H|1\rangle = |-\rangle$, $H|R\rangle = e^{i\pi/4}|L\rangle$, $H|L\rangle = e^{-i\pi/4}|R\rangle$, not a classical operation

- Phase operator P_θ:

$$P_\theta = |0\rangle\langle 0| + e^{i\theta}|1\rangle\langle 1| = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}$$

Properties: unitary, not Hermitian (except for $\theta = 0, \pm\pi, \ldots$), eigenvalues 1 with $|0\rangle$, $e^{i\theta}$ with $|1\rangle$, not a classical deterministic operation (except for $\theta = 0, \pm 2\pi, \ldots$)

Special cases: $P_0 = I$, $P_\pi = Z$,

$$S = P_{\pi/2} = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, \qquad T = P_{\pi/4} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}.$$

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Large information carrying systems can be constructed from smaller systems such as bits or qubits.

Systems consisting of *multiple smaller systems (components)* can always be viewed as forming a *signle compound system.*

All results of the preceding section apply directly to a compound system seen as whole.

However, a component of a compound system no longer behaves as if it was a single system.

If X and Y are classical systems with classical state sets Σ nad Γ, then the state set of the compound system consisting of X and Y is given by the Cartesian product of sets Σ and Γ,

$$\Sigma \times \Gamma = \{(a, b) : a \in \Sigma \text{ and } b \in \Gamma\}.$$

In ordering of the elements of the Cartesian product state sets we follow the "alphabetical" convention (significance decreasing left-to-right), e.g.,

$$\{a, b, c\} \times \{0, 1\} = \{(a, 0), (a, 1), (b, 0), (b, 1), (c, 0), (c, 1)\}.$$

Example: Probability vector in a system of classical bits X and Y

$$|\psi\rangle = \begin{pmatrix} p[(X, Y) = (0, 0)] \\ p[(X, Y) = (0, 1)] \\ p[(X, Y) = (1, 0)] \\ p[(X, Y) = (1, 1)] \end{pmatrix} = \begin{pmatrix} 1/2 \\ 0 \\ 0 \\ 1/2 \end{pmatrix}$$

From the above probabilistic state we can deduce that

$$p[X = a] = 1/2 \qquad \text{and} \qquad p[Y = b] = 1/2,$$

where $a = 0, 1$, and $b = 0, 1$. We also observe that

$$p[(X, Y) = (a, b)] \;\neq\; p[X = a]\, p[Y = b].$$

This property is known as *correlation*, or lack of independence of the components of the compound system.

Independence of two systems

Let X and Y be systems with classical state sets Σ and Γ.

If in the given probabilistic state of the system

$$p[(X, Y) = (a, b)] \;=\; p[X = a]\, p[Y = b]$$

holds for all $a \in \Sigma$ and $b \in \Gamma$, then the systems are said to be *independent.*

If in the given probabilistic state X and Y are independent, measuring the state of one system does not decrease the uncertainty of the state of the other.

The condition for the idependence of systems X and Y in the probabilistic state of (X, Y) given by

$$\sum_{(a,b)\in\Sigma\times\Gamma} p_{ab} \, |ab\rangle$$

is equivalent to the existence of probabilistic states

$$|\phi\rangle = \sum_{a\in\Sigma} q_a \, |a\rangle \,, \qquad |\psi\rangle = \sum_{b\in\Sigma} r_b \, |b\rangle \,,$$

such that $p_{ab} = q_a r_b$ for all $a \in \Sigma$ and $b \in \Gamma$.

If probabilistic states of X and Y are

$$|\phi\rangle = \sum_{a\in\Sigma} q_a \, |a\rangle \qquad \text{and} \qquad |\psi\rangle = \sum_{b\in\Sigma} r_b \, |b\rangle \,,$$

then the probabilistic state of (X, Y) can be given by the *tensor product*

$$|\pi\rangle = |\phi\rangle \otimes |\psi\rangle = \sum_{(a,b)\in\Sigma\times\Gamma} q_a r_b \, |ab\rangle \,,$$

or equivalently $\langle ab|\pi\rangle = \langle a|\phi\rangle \, \langle b|\psi\rangle$ for all $a \in \Sigma$ and $b \in \Gamma$, which is the condition for the independence of X and Y.

It follows that if the probabilistic state of (X, Y) can be written as a tensor product of the probabilistic states of X and Y, then X and Y are independent.

If the probabilistic state of (X, Y) is given by

$$|\Psi\rangle = \sum_{(a,b)\in\Sigma\times\Gamma} p_{ab}\,|ab\rangle\,,$$

and if both X and Y are measured, then the probability of obtaining the state $|ab\rangle$ is

$$p[(X, Y) = (a, b)] = \langle ab|\Psi\rangle = p_{ab}.$$

Let the state of (X, Y) be given by $\sum_{(a,b)\in\Sigma\times\Gamma} p_{ab}\,|ab\rangle$.

If only X is measured, and Y is *not* measured, then the probability of obtaining the state $|a\rangle$ is

$$p[X = a] = \sum_{b\in\Gamma} p[(X, Y) = (a, b)] = \sum_{b\in\Gamma} p_{ab}.$$

This type of measurement is known as *partial* measurement, and the resulting *reduced probabilistic state* of X alone is then

$$\sum_{a\in\Sigma} p[X = a]\,|a\rangle = \sum_{a\in\Sigma}\left(\sum_{b\in\Gamma} p_{ab}\right)|a\rangle\,.$$

Let the state of (X, Y) be given by $\sum_{(a,b)\in\Sigma\times\Gamma} p_{ab}\,|ab\rangle$.

Given that X has been measured in the particular state $|a\rangle$, the probability of measuring Y in the state $|b\rangle$ is given by the conditional probability

$$p[Y = b\,|\,X = a] = \frac{p[(X, Y) = (a, b)]}{p[X = a]} = \frac{p_{ab}}{\sum_{c\in\Gamma} p_{ac}}.$$

Therefore, given that X has been measured in the state $|a\rangle$, the probabilistic state of Y becomes updated to

$$\sum_{b\in\Gamma} p[Y = b\,|\,X = a]\,|b\rangle = \frac{1}{\sum_{c\in\Gamma} p_{ac}}\sum_{b\in\Gamma} p_{ab}\,|b\rangle$$

Deterministic and probabilistic operations on compound classical systems follow the scheme developed for single systems.

Example: Controlled NOT (aka Ctrl-X or CX) is a deterministic operation that performs a NOT operation on Y (target bit) if $X = 1$ (control bit) and does nothing otherwise,

$$|00\rangle\langle 00| + |01\rangle\langle 01| + |11\rangle\langle 10| + |10\rangle\langle 11| = \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & & 1 \\ & & 1 & \end{pmatrix}.$$

If X and Y exchange roles,

$$|00\rangle\langle 00| + |11\rangle\langle 01| + |10\rangle\langle 10| + |01\rangle\langle 11| = \begin{pmatrix} 1 & & & \\ & & & 1 \\ & & 1 & \\ & 1 & & \end{pmatrix}.$$

Example: Probabilistic operation that sets X to 0 or 1 with probabilities $1/2$ (regardless of the initial state of X), while it does nothing to Y,

$$\left(\frac{1}{2}|00\rangle + \frac{1}{2}|10\rangle\right)\left(\langle 00| + \langle 10|\right) + \left(\frac{1}{2}|01\rangle + \frac{1}{2}|11\rangle\right)\left(\langle 01| + \langle 11|\right) = \begin{pmatrix} 1/2 & & 1/2 & \\ & 1/2 & & 1/2 \\ 1/2 & & 1/2 & \\ & 1/2 & & 1/2 \end{pmatrix}.$$

Let M and N be probabilistic operations on X and Y given by

$$M = \sum_{a,b\in\Sigma} \alpha_{ab}\,|a\rangle\langle b|\,, \qquad N = \sum_{c,d\in\Gamma} \alpha_{cd}\,|c\rangle\langle d|\,.$$

If M and N operate on X and Y independently, the resulting operation on the compound system (X, Y) is equivalent to the operation of the tensor product

$$M \otimes N = \sum_{a,b\in\Sigma}\sum_{c,d\in\Gamma} \alpha_{ab}\alpha_{cd}\,|ac\rangle\langle bd|\,.$$

When acting on a product state, $M \otimes N$ has the property

$$(M \otimes N)(|\phi\rangle \otimes |\psi\rangle) = M\,|\phi\rangle \otimes N\,|\psi\rangle\,.$$

Example: Let M be the probabilistic operator that randomly sets the bit to 0 or 1, and let $N = I$ (does nothing). The matrix representation of the tensor product of the operators is

$$M \otimes N = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix} \otimes \begin{pmatrix} 1 & \\ & 1 \end{pmatrix} = \begin{pmatrix} 1/2 & & 1/2 & \\ & 1/2 & & 1/2 \\ 1/2 & & 1/2 & \\ & 1/2 & & 1/2 \end{pmatrix}.$$

The quantum state of the compound system consisting of components X and Y with classical state sets Σ and Γ can be written as

$$|\Psi\rangle = \sum_{(a,b)\in\Sigma\times\Gamma} \mu_{ab} |ab\rangle ,$$

where $\mu_{ab} \in \mathbb{C}$ and $|ab\rangle = |a\rangle \otimes |b\rangle$. If vectors

$$|\phi\rangle = \sum_{a\in\Sigma} \alpha_a |a\rangle \qquad \text{and} \qquad |\psi\rangle = \sum_{b\in\Gamma} \beta_b |b\rangle$$

exist such that

$$|\Psi\rangle = |\phi\rangle \otimes |\psi\rangle ,$$

then the state $|\Psi\rangle$ is said to be a *product state*. Otherwise, the state is said to be *entangled*.

Example: Entangled states of two qubits

Bell states:

$$|\phi^+\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$
$$|\phi^-\rangle = \frac{1}{\sqrt{2}}|00\rangle - \frac{1}{\sqrt{2}}|11\rangle$$
$$|\psi^+\rangle = \frac{1}{\sqrt{2}}|01\rangle + \frac{1}{\sqrt{2}}|10\rangle$$
$$|\psi^-\rangle = \frac{1}{\sqrt{2}}|01\rangle - \frac{1}{\sqrt{2}}|10\rangle$$

Bell states are orthonormal and comprise the so called Bell basis.

Example: Entangled states of three qubits

GHZ (Greenberger, Horne, Zeilinger) states:

$$\frac{1}{\sqrt{2}}|000\rangle + \frac{1}{\sqrt{2}}|111\rangle$$

W (Wolfgang Dür) state:

$$\frac{1}{\sqrt{3}}|001\rangle + \frac{1}{\sqrt{3}}|010\rangle + \frac{1}{\sqrt{3}}|100\rangle$$

If the quantum state of the compound system (X, Y) is $\Psi = \sum_{a,b} \mu_{ab} |ab\rangle$, then the probability of measuring the system in the state $|ab\rangle$ is $p_{ab} = |\mu_{ab}|^2$.

If only X is measured, then the probability of measuring the state a is

$$p_a = \sum_b p_{ab} = \sum_b |\mu_{ab}|^2.$$

However, unlike in the case of classical probabilistic states, the above result does not allow us to construct what would be "reduced quantum state vector" of the component X as in general no such vector exists.

Let the quantum state of (X, Y) be given by $\Psi = \sum_{a,b} \mu_{ab} |ab\rangle$.

Assuming that the state of X has been measured in the state $|a\rangle$, while the state of Y has not been measured, the state of the compound system is understood to "collapse" to the state

$$|a\rangle \otimes \frac{\sum_b \mu_{ab} |b\rangle}{\sqrt{\sum_b |\mu_{ab}|^2}}.$$

The above result is analogous to the result that was obtained in the context of classical probabilistic states.

Unitary operations on quantum states of compound systems follow the scheme developed for single systems.

Let the compound system (X, Y) consist of components X and Y. If operator M operates on X, and N independently operates on Y, then the joint operation is represented by the tensor product $M \otimes N$. If (X, Y) is in the product state, we have

$$(M \otimes N)(|\psi\rangle \otimes |\phi\rangle) = M|\psi\rangle \otimes N|\phi\rangle .$$

However, not every unitary operation on (X, Y) can be written as a tensor product of operations acting on X and Y.

Example: Hadamard operates on X, identity operates on Y,

$$H \otimes I = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix} \otimes \begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & & 1 & \\ & 1 & & 1\\ 1 & & -1 & \\ & 1 & & -1\end{pmatrix},$$

or the other way around,

$$I \otimes H = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} \otimes \begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1 & & \\ 1 & -1 & & \\ & & 1 & 1\\ & & 1 & -1\end{pmatrix}.$$

Example: If X and Y share the same classical state set, then the $SWAP$ operation is defined as $SWAP(|a\rangle \otimes |b\rangle) = |b\rangle \otimes |a\rangle$.

The matrix representing $SWAP$ operation follows from

$$SWAP = \sum_{a,b} |ba\rangle\langle ab| = \sum_{a,b} |b\rangle\langle a| \otimes |a\rangle\langle b| ,$$

which for qubits gives

$$\begin{pmatrix}1 & & & \\ & & 1 & \\ & 1 & & \\ & & & 1\end{pmatrix}.$$

Example: If Q is a qubit, R is an arbitrary system, and U_R is a unitary operator operating on R, then the controlled-unitary operation on (Q, R) is defined as

$$CU = |0\rangle\langle 0| \otimes I_R + |1\rangle\langle 1| \otimes U_R,$$

where the identity operator I_R and the unitary operator U_R operate on R. If R is a qubit, and $U_R = X$, one obtains the controlled-NOT operation,

$$CX = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X = \begin{pmatrix}1 & & & \\ & 1 & & \\ & & & 1\\ & & 1 & \end{pmatrix}.$$

Example: The Fredkin operation is defined in the system of three qubits as the controlled-SWAP operation,

$$CSWAP = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes SWAP,$$

therefore having the properties

$$CSWAP\,|0bc\rangle = |0bc\rangle , \qquad CSWAP\,|1bc\rangle = |1cb\rangle .$$

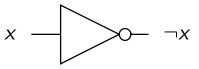
Example: The Toffoli operation is defined in the system of three qubits as the controlled-controlled-NOT operation,

$$CCX = (I - |11\rangle\langle 11|) \otimes I + |11\rangle\langle 11| \otimes X$$

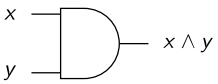
(note that $I - |11\rangle\langle 11| = |00\rangle\langle 00| + |01\rangle\langle 01| + |10\rangle\langle 10|$).

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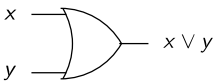
Deterministic operations NOT, AND, and OR, of classical logic can be represented by gate symbols and corresponding matrices:

NOT: 

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

AND: 

$$\begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

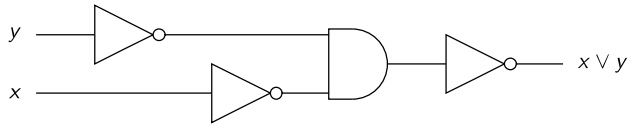
OR: 

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \end{pmatrix}$$

Complex logical operations in calssical logic can be represented by wire diagrams involving elementary logical gates. Classical logical operation of any level of complexity can be composed by using NOT and AND gates only (or by using NAND gate only).

Example: Creation of OR gate using NOT and AND gate

DeMorgan's identity: $x \vee y = \neg(\neg x \wedge \neg y)$



Matrix representation: $OR = NOT \cdot AND \cdot (NOT \otimes I) \cdot (I \otimes NOT)$

In general, the output of a known classical logical circuit does not allow one to reconstruct its input. This means that classical logic gates are in general irreversible. Any classical logic gate with more input wires that output wires, such as AND, OR, or NAND, is irreversible. If the number of classical gate's input wires equals the number of output wires, the gate may be reversible. NOT gate is reversible.

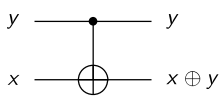
irreversibility of operations in classical logic implies that

- information is lost,
- entropy increases,
- energy is required for gates to operate,
- heat is generated (see Landauer's principle).

Irreversibility of certain classical gates also implies that these operations are not permissible within quantum information processing since time evolution within quantum mechanics is governed by unitary operators that are invertible.

However, it will be shown that all classical operations can be formulated reversibly and therefore used in quantum processing.

Controlled-NOT gate (CX):



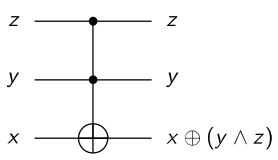
$$\begin{pmatrix} 1 & & & \\ & & & 1 \\ & & 1 & \\ & 1 & & \end{pmatrix}$$

Bit x is the target bit, bit y is the control bit.

Operator $\oplus$ is the binary XOR operator (mod 2 addition).

Proof of reversibility (CX is its own inverse): $CX \cdot CX = I$

Toffoli gate (CCX):



$$\begin{pmatrix} 1 & & & & & & & \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & & & & & 1 \\ & & & & 1 & & & \\ & & & & & 1 & & \\ & & & & & & 1 & \\ & & & 1 & & & & \end{pmatrix}$$

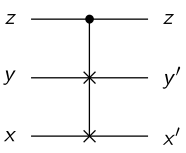
Bit x is the target bit, bits y and z are the two control bits.

Reversibility (CCX is its own inverse): $CCX \cdot CCX = I$

Universality of CCX (NOT and AND can be constructed):

$$(x, 1, 1) \to (\neg x, 1, 1) \qquad (0, y, z) \to (y \wedge z, y, z)$$

Fredkin gate (*CSWAP*):



$$\begin{pmatrix} 1 & & & & & & & \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & 1 & & & & \\ & & & & 1 & & & \\ & & & & & 1 & & \\ & & & & & & 1 & \\ & & & & & & & 1 \end{pmatrix}$$

Bit z is the control bit:

$$(x, y, 0) \to (x, y, 0) \qquad (x, y, 1) \to (y, x, 1)$$

Reversibility: $CSWAP \cdot CSWAP = I$

Universality:

$$(0, 1, z) \to (z, \neg z, z) \qquad (x, 0, z) \to ((\neg z) \wedge x, x \wedge z, z)$$

Existence of universal reversible gates (Toffoli and Fredkin gates) implies that any classical logical circuit ca be, in principle, formulated as a reversible logical circuit.

Classical computer is a system of classical bits. n-bit classical computer can be found in any of its $N = 2^n$ classical states.

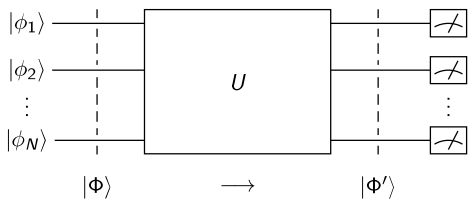
Quantum computer is a system of quantum bits or qubits. The state of n-qubit quantum computer is any unit-normalized linear combination (superposition) of the $N = 2^n$ orthogonal basis states spanning the Hilbert space $\mathcal{H}^{\otimes n}$.

Measurement of the state of the classical and of the quantum computer have vastly different nature:

- Classical computer can be in any of the $N = 2^n$ classical states. Measurement reveals the state the computer is in. The state of the computer is not affected by the measurement process.
- Quantum computer can be in any of infinitely many different states (superpositions of $N = 2^n$ basis states). The standard basis measurement of its state gives one of the basis states and it is assumed that this is the state the computer is in after the measurement. This implies that the measurement process in general changes the state of the system.

Evolution of the state of a quantum computer is described by a unitary transformation U:

$$|\Phi\rangle \to |\Phi'\rangle = U\,|\Phi\rangle$$



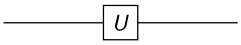
Unitarity of time evolution implies the existence of the inverse operation, $U^{-1} = U^\dagger$, which implies the *reversibility* of computation at the quantum level (before measurement).

Since every classical algorithm can, using reversible classical gates, be formulated as a reversible classical logical circuit, it can, at least in principle, also be understood as a quantum logical circuit and be executed on a quantum computer.

Apart form classical algorithms, quantum computers can execute logical circuits involving gates not available within classical logic. Such quantum algorithms may have properties fundamentally different from those of classical algorithms.

A quantum gate is any unitary operation operating on one or several qubits.

The gate U operating on one qubit is shown with the symbol

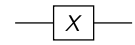


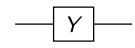
The gate is defined with

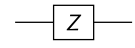
$$U|0\rangle = \alpha_{00}|0\rangle + \alpha_{10}|1\rangle$$
$$U|1\rangle = \alpha_{01}|0\rangle + \alpha_{11}|1\rangle$$

where $U = \begin{pmatrix} \alpha_{00} & \alpha_{01} \\ \alpha_{10} & \alpha_{11} \end{pmatrix}$ is a unitary matrix ($U^\dagger \cdot U = I$).

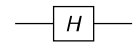
Example: Pauli gates (operators, matrices)

$$X = \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$


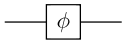
$$Y = \sigma_y = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}$$


$$Z = \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$


Example: Hadamard operator

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$


Example: Phase operator

$$P[\phi] = \begin{pmatrix} 1 & 0 \\ 0 & \mathrm{e}^{\mathrm{i}\phi} \end{pmatrix}$$


rotates the state about the z-axis. Special cases are

$$S = P[\pi/2] = \begin{pmatrix} 1 & 0 \\ 0 & \mathrm{i} \end{pmatrix} \quad \text{and} \quad T = P[\pi/4] = \begin{pmatrix} 1 & 0 \\ 0 & \mathrm{e}^{\mathrm{i}\pi/4} \end{pmatrix}$$

Rotations about x and y-axes can be achieved by

$$P_x[\phi] = H \cdot P[\phi] \cdot H \quad \text{and} \quad P_y[\phi] = T \cdot H \cdot P[\phi] \cdot H \cdot T^{-1}.$$

Example: Square-root-of-NOT operator is defined as

$$\sqrt{\mathrm{NOT}} = \sqrt{X} = \frac{1}{1+\mathrm{i}} \begin{pmatrix} 1 & \mathrm{i} \\ \mathrm{i} & 1 \end{pmatrix}.$$

The matrix representing $\sqrt{X}$ is unitary,

$$\left(\sqrt{X}\right)^\dagger \cdot \sqrt{X} = I,$$

and the name of the operator is due to the property

$$\sqrt{X} \cdot \sqrt{X} = X.$$

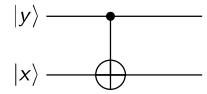
Useful identity:

$$\sqrt{X} = H \cdot S \cdot H$$

The general quantum logic gate $\mathcal{H}^{\otimes n}$ can be represented by a matrix of dimension $2^n \times 2^n$.

Theorem (no proof here): The general unitary transformation in $\mathcal{H}^{\otimes n}$ can be constructed using controlled-NOT (CX) gates and one-qubit unitary transformations.

The controlled-NOT (CX) gate is inherited from classical logic:



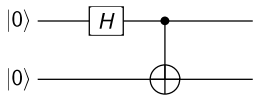
If x and y are classical bit values,

$$|x\rangle \otimes |y\rangle \to |x \oplus y\rangle \otimes |y\rangle$$

Example: Let the state of a two-qubit system initially be a product state

$$|\Phi\rangle = |0\rangle \otimes |0\rangle = |00\rangle\,.$$

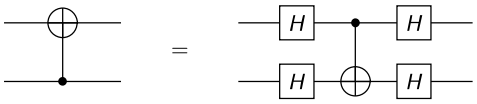
At the exit of the quantum logical circuit



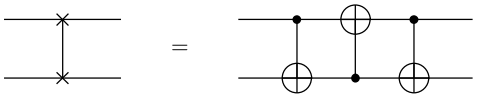
one obtains the entangled state

$$|\Phi'\rangle = \frac{1}{\sqrt{2}}(\,|00\rangle + |11\rangle\,).$$

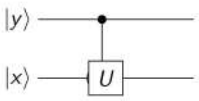
Example: Inverted CX gate identity



Example: Construction of $SWAP$ using CX gates



The controlled-unitary (CU) gate,



Qubit $|x\rangle$ is the target qubit, qubit $|y\rangle$ is the control qubit.

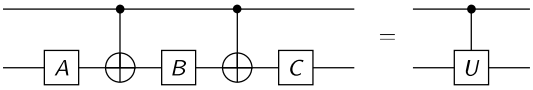
- If $|y\rangle = |0\rangle$, then $|x\rangle$ remains unchanged,
- while if $|y\rangle = |1\rangle$, then $|x\rangle \to U\,|y\rangle$.

With $U = X$, this becomes the CX gate.

Example: If operators A, B, and C, satisfy

$$CBA = I, \qquad CXBXA = U,$$

then the following identity holds:



Proof using matrix representation of operators:

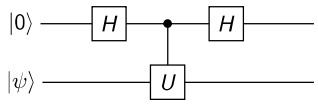
$$\begin{pmatrix} C & 0 \\ 0 & C \end{pmatrix} \cdot \begin{pmatrix} I & 0 \\ 0 & X \end{pmatrix} \cdot \begin{pmatrix} B & 0 \\ 0 & B \end{pmatrix} \cdot \begin{pmatrix} I & 0 \\ 0 & X \end{pmatrix} \cdot \begin{pmatrix} A & 0 \\ 0 & A \end{pmatrix}$$

$$= \begin{pmatrix} CBA & 0 \\ 0 & CXBXA \end{pmatrix} = \begin{pmatrix} I & 0 \\ 0 & U \end{pmatrix}$$

Example: Let $|\psi\rangle$ be an eigenstate of the unitary operator U with the eigenvalue λ,

$$U\,|\psi\rangle = \lambda\,|\psi\rangle\,, \qquad |\lambda| = 1.$$

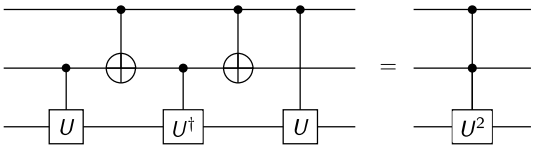
At the exit of the quantum logical circuit



one obtains the state

$$|\psi\rangle \otimes \left(\frac{1+\lambda}{2}\,|0\rangle + \frac{1-\lambda}{2}\,|1\rangle\right).$$

Example: If U is a unitary operator, then the following identity among the circuits holds:



Toffoli gate is obtained with

$$U = \sqrt{X} = H \cdot S \cdot H$$

Single systems
Classical systems: Probabilistic states and operations
Quantum systems: Unitary operations
Single qubit states and operators
Multiple systems
Classical systems
Quantum systems
Logical circuits
Reversible formulation of classical logic
Quantum logical circuits
4 Entanglement

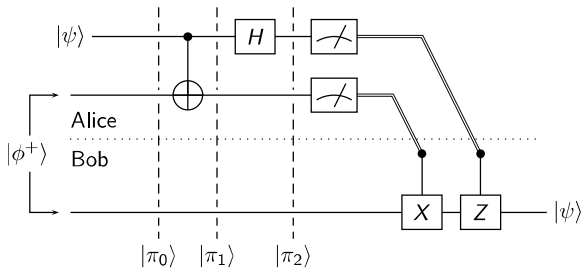
Quantum teleportation is a communication protocol that allows Alice to transfer an *unknown state* of a qubit to Bob by communicating only *two classical bit values.*

However, the protocol requires a pair of entangled qubits to be pre-shared between Alice and Bob.

Entanglement is in this context considered a computational resource.

A pair of entangled qubits represents the smallest quantity of entanglement and is called *ebit* (a bit of entanglement).

The entangled pair of qubits in the Bell state $|\phi^+\rangle$ is pre-shared between Alice and Bob. The state $|\psi\rangle$ of the third qubit is unknown to A and B.



The unknown state is teleported to Bob's end.

Assuming the unknown state can be given by the state vector

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

the initial state of the system is

$$\begin{aligned}|\pi_0\rangle &= |\phi^+\rangle \otimes |\psi\rangle \\ &= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \otimes (\alpha|0\rangle + \beta|1\rangle) \\ &= \frac{1}{\sqrt{2}}(\alpha|000\rangle + \beta|001\rangle + \alpha|110\rangle + \beta|111\rangle).\end{aligned}$$

After the operation of the cX gate the state is

$$\begin{aligned}|\pi_1\rangle &= \frac{1}{\sqrt{2}}(\alpha|000\rangle + \beta|011\rangle + \alpha|110\rangle + \beta|101\rangle) \\ &= \frac{\alpha}{\sqrt{2}}(|00\rangle + |11\rangle) \otimes |0\rangle + \frac{\beta}{\sqrt{2}}(|01\rangle + |10\rangle) \otimes |1\rangle,\end{aligned}$$

and after the Hadamard gate we have

$$\begin{aligned}|\pi_2\rangle &= \frac{\alpha}{\sqrt{2}}(|00\rangle + |11\rangle) \otimes H|0\rangle + \frac{\beta}{\sqrt{2}}(|01\rangle + |10\rangle) \otimes H|1\rangle \\ &= \frac{\alpha}{2}(|00\rangle + |11\rangle) \otimes (|0\rangle + |1\rangle) + \frac{\beta}{2}(|01\rangle + |10\rangle) \otimes (|0\rangle - |1\rangle).\end{aligned}$$

The state $|\pi_2\rangle$ can be factored as

$$\begin{aligned}|\pi_2\rangle = &\frac{1}{2}(\alpha|0\rangle + \beta|1\rangle) \otimes |00\rangle \\ &+ \frac{1}{2}(\alpha|0\rangle - \beta|1\rangle) \otimes |01\rangle \\ &+ \frac{1}{2}(\alpha|1\rangle + \beta|0\rangle) \otimes |10\rangle \\ &+ \frac{1}{2}(\alpha|1\rangle - \beta|0\rangle) \otimes |11\rangle,\end{aligned}$$

which distinguishes the states of Bob's qubit corresponding to the four possible outcomes of the partial standard basis measurement performed by Alice on her pair of qubits.

For example, assuming Alice has measured $|11\rangle$, the state of Bob's qubit collapses to

$$\alpha|1\rangle - \beta|0\rangle .$$

In this case Alice sends to Bob the classical value 1 for both bits, so the controlled operators X and Z both operate. The final state of Bob's qubit becomes

$$ZX(\alpha|1\rangle - \beta|0\rangle) = Z(\alpha|0\rangle - \beta|1\rangle) = \alpha|0\rangle + \beta|1\rangle ,$$

which is exactly the unknown state $|\psi\rangle$.

The remaining three cases can be worked out analogously.

Notes on quantum teleportation:

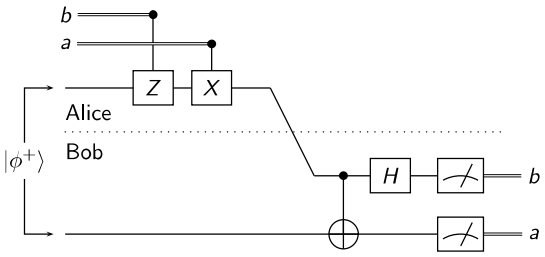
- Teleportation scheme does not violate the no-clonng theorem as the quantum state that Alice teleports to Bob is no longer present at Alice's end.
- In general, the unknown state being teleported may be entangled with further quantum systems, and therefore not be representable by a state vector as was assumed in our simplified analysis. However, it can be shown that the teleportation scheme presented works right in that case as well.

Superdense coding is a communication protocol that enables Alice to send two classical bit values to Bob by sending a single qubit.

This protocol is, in a sense, complementary to quantum teleportation.

Superdense coding relies on one ebit (pair of entangled qubits) being preshared between Alice and Bob.

Superdense coding logical circuit:



It can be shown that the ebit initialized in the Bell state $|\phi^+\rangle$, depending on the choice of the bit values a and b controlling the operators Z and X at Alice's end, brings the ebit into one of the four Bell states.

The circuit at Bob's end consisting of the cX and H operators is exactly the inverse of the familiar code used to generate Bell states.

The classical bit values a and b are therefore restored by standard basis measurements at Bob's end.

The CHSH game originates from the experiment proposed by John Clauser, Michael Horne, Abner Shimony, and Richard Holt, in their 1969 paper. The experiment was designed to demonstrate the role of entanglement in a verifyable way.

The idea behind the experiment is often presented as a game, since within game theory it falls into the category of non-local games.

The CHSH experiment can also be seen as a possible Bell test, i.e. a test case for the Bell inequalities, named after John Bell, that demonstrate the statistical nature of quantum correlations that do not exist in classical systems.

A non-local game is run by a referee and the players Alice and Bob who cooperate to win the game, but are not allowed to communicate during the game. The referee asks questions and obtains answers from Alice and Bob. Alice (Bob) does not know what question Bob (Alice) is being asked, nor what his (her) answer is.

The game is defined by a strict set of rules known to the referee, Alice and Bob. In one run of the game, Alice answers her question, Bob answers his question, and depending on their answers, they either win or lose the game.

For any given set of rules defining a non-local game, Alice and Bob are expected to develop a strategy maximizing the probability to win the game.

The rules of the CHSH game:

The questions and the answers are bits. The question given to Alice is $x \in \{0,1\}$ and her answer is $a \in \{0,1\}$. The question given to Bob is $y \in \{0,1\}$ and his answer is $b \in \{0,1\}$. Alice and Bob win if

$$a \oplus b = x \wedge y$$

and lose otherwise, i.e. the four cases of the game that appear randomly with uniform probability are

(x,y)	win	lose
$(0,0)$	$a = b$	$a \neq b$
$(0,1)$	$a = b$	$a \neq b$
$(1,0)$	$a = b$	$a \neq b$
$(1,1)$	$a \neq b$	$a = b$

Deterministic (and probabilistic) strategies:

There is no deterministic strategy that wins with probability larger than 3/4.

The proof requires considering all 16 deterministic strategies individually, or an argument as follows: Assuming Alice and Bob win for $(x,y) = (0,0)$ implies $a(0) = b(0)$. If they also win for $(x,y) = (0,1)$ then $b(1) = a(0)$, and if they win for $(x,y) = (1,0)$ then $a(1) = b(0)$. Therefore, if they win in these three cases their strategy is $a(0) = b(0) = a(1) = b(1)$. This implies they lose in the fourh case $(x,y) = (1,1)$.

A probabilistic strategy need not be considered as it can't do any better than the best deterministic strategy.

In order to formulate the quantum strategy for the CHSH game we introduce the orthogonal states

$$|\theta\rangle = \cos\theta\,|0\rangle + \sin\theta\,|1\rangle\,, \qquad |\theta_\perp\rangle = -\sin\theta\,|0\rangle + \cos\theta\,|1\rangle\,,$$

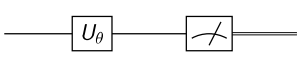
We also introduce the unitary operator

$$U_\theta = |0\rangle\langle\theta| + |1\rangle\langle\theta_\perp|$$

with the properties

$$U_\theta\,|0\rangle = Z\,|\theta\rangle \qquad U_\theta\,|1\rangle = -Z\,|\theta_\perp\rangle \qquad U_{\theta'}U_\theta = U_{\theta'+\theta}$$

Consider U_θ operating before the measurement gate,



The effect of U_θ is that the state $|\theta\rangle$ is measured as 0, and $|\theta_\perp\rangle$ is measured as 1. This application of U_θ can be understood as the "rotation" of the standard measurement basis by the angle θ, and is known as the measurement in the U_θ basis.

If both qubits of the Bell state $|\phi^+\rangle$ are to be measured in the U_θ basis, the operation of U_θ gates gives

$$\begin{aligned}(U_\theta \otimes U_\theta)\,|\phi^+\rangle &= \frac{1}{\sqrt{2}}\left(U_\theta\,|0\rangle \otimes U_\theta\,|0\rangle + U_\theta\,|1\rangle \otimes U_\theta\,|1\rangle\right) \\ &= \frac{1}{\sqrt{2}}\left(Z\,|\theta\rangle \otimes Z\,|\theta\rangle + Z\,|\theta_\perp\rangle \otimes Z\,|\theta_\perp\rangle\right) \\ &= \cdots = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right),\end{aligned}$$

which means that Alice and Bob are certain to measure $a = b$ with all rotations of the measurement basis.

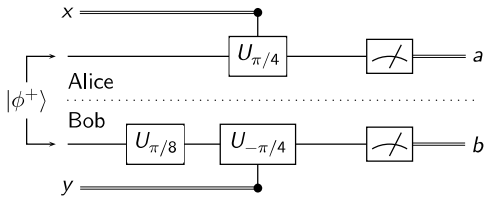
If Alice and Bob use measurement bases that are different, then

$$\begin{aligned}(U_\theta \otimes I)\,|\phi^+\rangle &= \frac{1}{\sqrt{2}}\left(U_\theta\,|0\rangle \otimes I\,|0\rangle + U_\theta\,|1\rangle \otimes I\,|1\rangle\right)\\ &= \frac{1}{\sqrt{2}}\left(Z\,|\theta\rangle \otimes |0\rangle - Z\,|\theta_\perp\rangle \otimes |1\rangle\right)\\ &= \cdots = \frac{1}{\sqrt{2}}\left(c\,|00\rangle + s\,|01\rangle - s\,|10\rangle + c\,|11\rangle\right),\end{aligned}$$

where $c = \cos\theta$, $s = \sin\theta$. This means that the probabilities of Alice and Bob measuring $a = b$ and $a \neq b$ are

$$p(a = b) = \cos^2\theta, \qquad p(a \neq b) = \sin^2\theta.$$

The circuit representing the strategy:



The probability of winning the CHSH game turns out to be

$$\left|\cos\frac{\pi}{8}\right|^2 = \frac{2+\sqrt{2}}{4} \simeq 0.85$$

in all four cases, beating the deterministic strategy.

The result on the preceding slide can be confirmed by direct calculation or by a geometric argument as follows.

Unitary operators preceding the measurement gates determine the relative orientation $|\theta_B - \theta_A|$ of the effective measurement bases that Alice and Bob are using.

(x, y)	θ_A	θ_B	$\lvert\theta_B - \theta_A\rvert$
$(0,0)$	0	$\pi/8$	$\pi/8$
$(0,1)$	0	$-\pi/8$	$\pi/8$
$(1,0)$	$\pi/4$	$\pi/8$	$\pi/8$
$(1,1)$	$\pi/4$	$-\pi/8$	$3\pi/8$

The probability of obtaining $a = b$ is given by $|\cos(\theta_B - \theta_A)|^2$, which confirms the result.

Deutsch's algorithm

Quantum computers (SI)

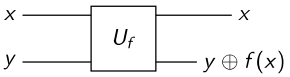
January 19, 2024

- If the initial state of a quantum computer is a superposition of computational basis states, then the final state of the computer is the superposition of the final states corresponding to computational basis states on the input.
- This implies that the output state of a quantum computer may contain some information (in superposition) on the output states corresponding to various individual inputs.
- Quantum algorithms may exploit this oportunity and solve some computational problems more efficiently (e.g. evaulate a function, or run a code, not so many times) than the purely classical counterparts. This principle is colloquially known as *quantum parallelism*.

On a classical computer, a function $f : \{0,1\} \to \{0,1\}$ can be implemented reversibly by the transformation U_f:

$$(x, y) \xrightarrow{U_f} (x, y \oplus f(x))$$

Ligical circuit:



The upper bit is (sometimes) called the input bit, the lower bit is the output bit.

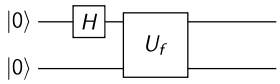
Reversibility is obvious since $U_f^2 = I$:

$$(x, y) \xrightarrow{U_f^2} (x, (y \oplus f(x)) \oplus f(x)) = (x, y)$$

A quantum gate representing $f : \{0,1\} \to \{0,1\}$, a.k.a. *quantum oracle*, has the property

$$U_f \left|y \otimes x\right\rangle = \left|(y \oplus f(x)) \otimes x\right\rangle, \qquad (x, y \in \{0,1\}).$$

Example: At the exit of the quantum circuit which evaluates U_f only once,



the state of the system

$$\frac{1}{\sqrt{2}}\big(\left|f(0) \otimes 0\right\rangle + \left|f(1) \otimes 1\right\rangle\big).$$

involves the function values for both values of its argument.

Generalization to n input and m output qubits

- The n-qubit input register carries the argument of the function. It's states can be represented using the basis

  $$\{\left|x\right\rangle ; x = 0, \ldots, 2^n - 1\},$$

  or $\left|x\right\rangle = \left|x_{n-1} \cdots x_1 x_0\right\rangle$, where $x_{n-1}, \ldots, x_1, x_0$ are bit values.
- The m-qubit output register carries the function value. It's states can be represented using the basis

  $$\{\left|z\right\rangle ; z = 0, \ldots, 2^m - 1\},$$

  or $\left|z\right\rangle = \left|z_{m-1} \cdots z_1 z_0\right\rangle$, where $x_{n-1}, \ldots, x_1, x_0$ are bit values.

- The Hadamard operator acting on a register is a tensor product of Hadamard operators operating on all qubits. Operating on the input register initialized to $\left|0^{\otimes n}\right\rangle = \left|0\right\rangle \otimes \cdots \otimes \left|0\right\rangle$ it gives

  $$H^{\otimes n} \left|0^{\otimes n}\right\rangle = \frac{1}{2^{n/2}} \sum_{x=0}^{2^n-1} \left|x\right\rangle.$$

- The gate U_f implementing the function $f(x)$ is defined as

  $$U_f \left|z \otimes x\right\rangle = \left|(z \oplus f(x)) \otimes x\right\rangle,$$

  where $\oplus$ is mod-2 addition without carryover (*bitwise*). Unitarity of U_f follows from the property $U_f^2 = I$.

- It can be shown that $U_f \left|0^{\otimes m} \otimes x\right\rangle = \left|f(x) \otimes x\right\rangle$.
- The input register is initialized to $\left|0^{\otimes n}\right\rangle$ and is operated on by $H^{\otimes n}$. The output register is initialized to $\left|0^{\otimes m}\right\rangle$. At the output of U_f we have the state

  $$\begin{aligned} U_f\left(\left|0^{\otimes m}\right\rangle \otimes H^{\otimes n}\left|0^{\otimes n}\right\rangle\right) &= \frac{1}{2^{n/2}} \sum_{x=0}^{2^n-1} U_f \left|0^{\otimes m} \otimes x\right\rangle \\ &= \frac{1}{2^{n/2}} \sum_{x=0}^{2^n-1} \left|f(x) \otimes x\right\rangle \end{aligned}$$

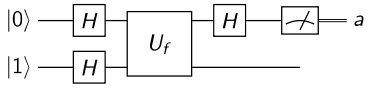
  that contains information of the function values for all its 2^n argument values.

Detusch's algorithm:

The algorithm determines whether $f : \{0,1\} \to \{0,1\}$ is "balanced", $f(1) \neq f(0)$, or it is "constant", $f(1) = f(0)$.

Classicaly, two calls to f are required to compute $f(0)$ and $f(1)$, which are then compared.

Deutsch's algorithm calls the quantum implementation of f only once:



With balanced f we get $a = 0$, while with constant f we get $a = 1$.

The two-qubit system is initialized in the state

$$\left|\pi_0\right\rangle = \left|1\right\rangle \otimes \left|0\right\rangle.$$

After the pair of Haramard gates this becomes

$$\left|\pi_1\right\rangle = (H \otimes H)(\left|1\right\rangle \otimes \left|0\right\rangle) = H\left|1\right\rangle \otimes H\left|0\right\rangle = \left|-\right\rangle \otimes \left|+\right\rangle,$$

which can we written as

$$\left|\pi_1\right\rangle = \frac{1}{2}(\left|0\right\rangle - \left|1\right\rangle) \otimes \left|0\right\rangle + \frac{1}{2}(\left|0\right\rangle - \left|1\right\rangle) \otimes \left|1\right\rangle.$$

After the U_f gate the state can be written as

$$|\pi_2\rangle = \frac{1}{2}\big(\,|0\oplus f(0)\rangle - |1\oplus f(0)\rangle\,\big)\otimes|0\rangle + \frac{1}{2}\big(\,|0\oplus f(1)\rangle - |1\oplus f(1)\rangle\,\big)\otimes|1\rangle\,.$$

Using the formula $|0\oplus x\rangle - |1\oplus x\rangle = (-1)^x(|0\rangle - |1\rangle)$ valid for $x = 0, 1$ the above state can be written as

$$\begin{aligned}|\pi_2\rangle &= \frac{1}{2}(-1)^{f(0)}(|0\rangle - |1\rangle)\otimes|0\rangle + \frac{1}{2}(-1)^{f(1)}(|0\rangle - |1\rangle)\otimes|1\rangle \\ &= \frac{1}{2}(|0\rangle - |1\rangle)\otimes\big((-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle\,\big) \\ &= \frac{1}{2}(-1)^{f(0)}(|0\rangle - |1\rangle)\otimes\big(\,|0\rangle + (-1)^{f(1)-f(0)}|1\rangle\,\big),\end{aligned}$$

In the last expression (preceding slide) we recognize the two cases:

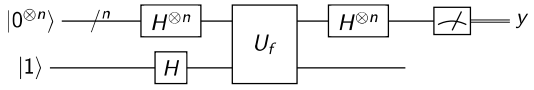
$$|\pi_2\rangle = \begin{cases}(-1)^{f(a)}\,|-\rangle\otimes|+\rangle & \text{if} \quad f(0) = f(1)\\ (-1)^{f(a)}\,|-\rangle\otimes|-\rangle & \text{if} \quad f(0)\neq f(1)\end{cases}$$

After the last Hadamard gate on the input qubit this becomes:

$$|\pi_3\rangle = \begin{cases}(-1)^{f(a)}\,|-\rangle\otimes|0\rangle & \text{if} \quad f(0) = f(1)\\ (-1)^{f(a)}\,|-\rangle\otimes|1\rangle & \text{if} \quad f(0)\neq f(1)\end{cases}$$

Deutsch–Jozsa algorithm:

Given the promise that $f : \{0,1\}^n \to \{0,1\}$ is either constant or balanced, D–J algorithm outputs 0 if f is constant, and outputs 1 if f is balanced.



Technically, if all n qubits of the input register are measured in the state 0, then f is constant. Otherwise, f is balanced.

A deterministic classical algorithm may require up to $2^{n-1}+1$ evaluations of f to solve the D–J problem. (However, efficient probabilistic classical algorithms are easy to formulate.)

As a preliminary for the analysis of the D–J algorithm we use the formula

$$H\,|a\rangle = \frac{1}{\sqrt{2}}\big(\,|0\rangle + (-1)^a\,|1\rangle\,\big) = \frac{1}{\sqrt{2}}\sum_{b\in\{0,1\}}(-1)^{ab}\,|b\rangle$$

which is valid for $a \in \{0,1\}$ to express the operation of $H^{\otimes n}$ on a n-qubit register as

$$\begin{aligned}&H^{\otimes n}\,|x_{n-1}\cdots x_0\rangle = H\,|x_{n-1}\rangle\otimes\cdots\otimes H\,|x_0\rangle = \\ &= \frac{1}{\sqrt{2}}\sum_{y_{n-1}\in\{0,1\}}(-1)^{x_{n-1}y_{n-1}}\,|y_{n-1}\rangle\otimes\cdots\otimes\frac{1}{\sqrt{2}}\sum_{y_0\in\{0,1\}}(-1)^{x_0y_0}\,|y_0\rangle \\ &= \frac{1}{\sqrt{2^n}}\sum_{y_{n-1}\cdots y_0\in\{0,1\}^n}(-1)^{x_{n-1}y_{n-1}+\cdots+x_0y_0}\,|y_{n-1}\cdots y_0\rangle\,.\end{aligned}$$

The system is initialized in the state

$$|1\rangle\otimes|0^{\otimes n}\rangle\,,$$

and after the first Hadamard gates we have (using the formula from the preceding slide)

$$(H\otimes H^{\otimes n})(|1\rangle\otimes|0^{\otimes n}\rangle) = |-\rangle\otimes\frac{1}{\sqrt{2^n}}\sum_{x_{n-1}\cdots x_0\in\{0,1\}^n}|x_{n-1}\cdots x_0\rangle\,.$$

After the U_f operator the state becomes

$$|-\rangle\otimes\frac{1}{\sqrt{2^n}}\sum_x(-1)^{f(x_{n-1}\cdots x_0)}\,|x_{n-1}\cdots x_0\rangle\,.$$

After last Haramard gates the state of the input register is

$$\frac{1}{2^n}\sum_x(-1)^{f(x)}\sum_y(-1)^{x_{n-1}y_{n-1}+\cdots+x_0y_0}\,|y_{n-1}\cdots y_0\rangle\,.$$

Now the probability of measuring the register in the state $|0^{\otimes n}\rangle$ is

$$\left|\frac{1}{2^n}\sum_x(-1)^{f(x)}\right|^2,$$

which is 1 for a constant function and is 0 for a balanced function.

Grover's algorithm

Kvantna računala (SI)

20. siječnja 2024.

We are considering the search problem in a unstructured (i.e. unsorted) database.

Let $f : \{0, \ldots, 2^n - 1\} \to \{0, 1\}$ have the property

$$f(x) = \delta_{xw} = \begin{cases} 0 & \text{for} \quad x \neq w \\ 1 & \text{for} \quad x = w \end{cases}$$

where w is the unknown number to be determined (the "winner").

In order to determine w, a classical algorithm must evaluate f on the average $2^n/2$ times.

The quantum implementation of $f(x) = \delta_{xw}$ is achieved by a n-qubit unitary operator U_f such that

$$U_f |x\rangle = (-1)^{f(x)} |x\rangle , \qquad x = 0, \ldots, 2^n - 1.$$

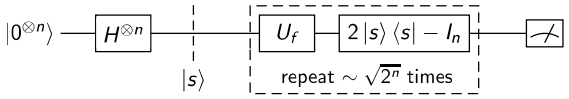
Unitarity of U_f follows from the property $U_f^2 = I$.

The operator U_f can be expressed as

$$U_f = I - 2 |w\rangle\langle w|$$

Operators (functions) that can be queried (evaluated) as many times as is needed, but whose internal workings are hidden from the main algorithm that is attempting to infer their properties, are called *black boxes* or *quantum oracles*.

The logical circuit of Grover's algorithm:



The operator in the dashed box is known as Grover's operator. This operator is repeated approximately $\sqrt{2^n}$ times, which is exponentially less than the number of evaluations of the function f required by the classical search algorithm.

The final state of the system is close to the required state $|w\rangle$.

Analysis of Grover's algorithm:

- All n qubits are initialized in the state $|0\rangle$.
- Hadamard operators operating on all n qubits brings the system into the uniform superposition $|s\rangle$ of the 2^n standard basis states with real positive coefficients,

$$|s\rangle = H^{\otimes n} |0^{\otimes n}\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle .$$

- Owing to all cofficients being real, the analysis of the Grover's algorithm can be carried out in the Euklid vector space of dimension 2^n.

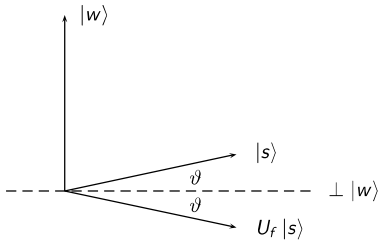
- Observe that, for large n, the state $|s\rangle$ is "almost orthogonal" to any of the states $|x\rangle$ of the standard basis,

$$\langle x|s\rangle = \frac{1}{\sqrt{2^n}}.$$

- Introduce the hyperplane (subspace) of dimension $2^n - 1$ orthogonal to $|w\rangle$. (Vectors orthogonal to $|w\rangle$ belong to the hyperplane, while $|w\rangle$ is orthogonal to the hyperplane.)
- Introduce the angle ϑ that $|s\rangle$ subtends with any of the vectors in the hyperplane,

$$\langle w|s\rangle = \frac{1}{\sqrt{2^n}} = \sin\vartheta.$$

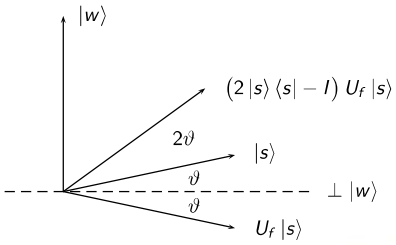
- The operation of U_f on a general state flips the sign of the coefficient of the basis vector $|w\rangle$.
- Representing $|s\rangle$ as a vector in the plane spanned by $|w\rangle$ and $|s\rangle$, the operation of U_f can be understood as the reflection of $|s\rangle$ in the axis perpendicular to $|w\rangle$:



- The operator

$$2|s\rangle\langle s| - I = |s\rangle\langle s| - (I - |s\rangle\langle s|)$$

can be understood as reflecting the state in the axis determined by the vector $|s\rangle$. The figure shows the states $|s\rangle$, $U_f|s\rangle$, and $(2|s\rangle\langle s| - I)U_f|s\rangle$:



- As the composition of two reflections in axes subtending an angle α is equivalent to a rotation for 2α, the operation

$$(2|s\rangle\langle s| - I)U_f$$

can be understood as a rotation by

$$2\vartheta.$$

- Initial state $|s\rangle$ subtends the angle ϑ with the axis perpendicular to $|w\rangle$, so after k operations of $(2|s\rangle\langle s| - I)U_f$ it subtends the angle

$$(2k+1)\,\vartheta.$$

- The operator $(2|s\rangle\langle s| - I)U_f$ is applied as many times as is needed to bring the initial state $|s\rangle$ as close as possible to the state $|w\rangle$. The requirement

$$(2k+1)\vartheta \simeq \pi/2,$$

gives

$$k \simeq \sqrt{2^n}.$$

We've shown that Grover's quantum algorithm finds $|w\rangle$ with large probability after evaluating f approximately $\sqrt{2^n}$ times, while the classical algorithm requires $2^n/2$ evaluations on the average. As the number of eveluations required by the classical algorithm is quadratic in the number of evaluations needed by the quantum algorithm, the quantum algorithm presents a quadratic speedup relative to the classical one.